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(54) **METHOD FOR HYPOTHERMIC TRANSPORT OF BIOLOGICAL SAMPLES**

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(58) **Field of Classification Search**
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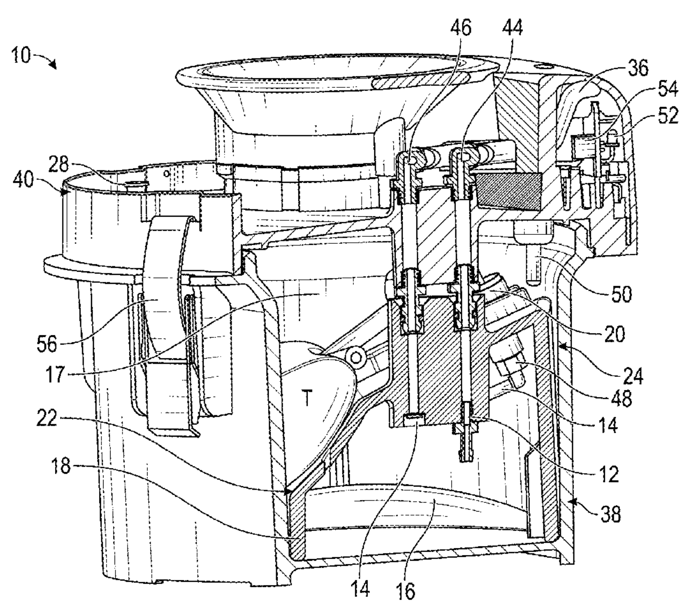
(57) **ABSTRACT**

A method for transporting a biological sample at hypothermic temperatures. The container can be cooled using phase change material and pump fluid through the sample. The fluid can pump through the system at a rate independent of the parameters of the biological sample. A valve can control the rate of flow of the fluid into the biological sample.

18 Claims, 20 Drawing Sheets



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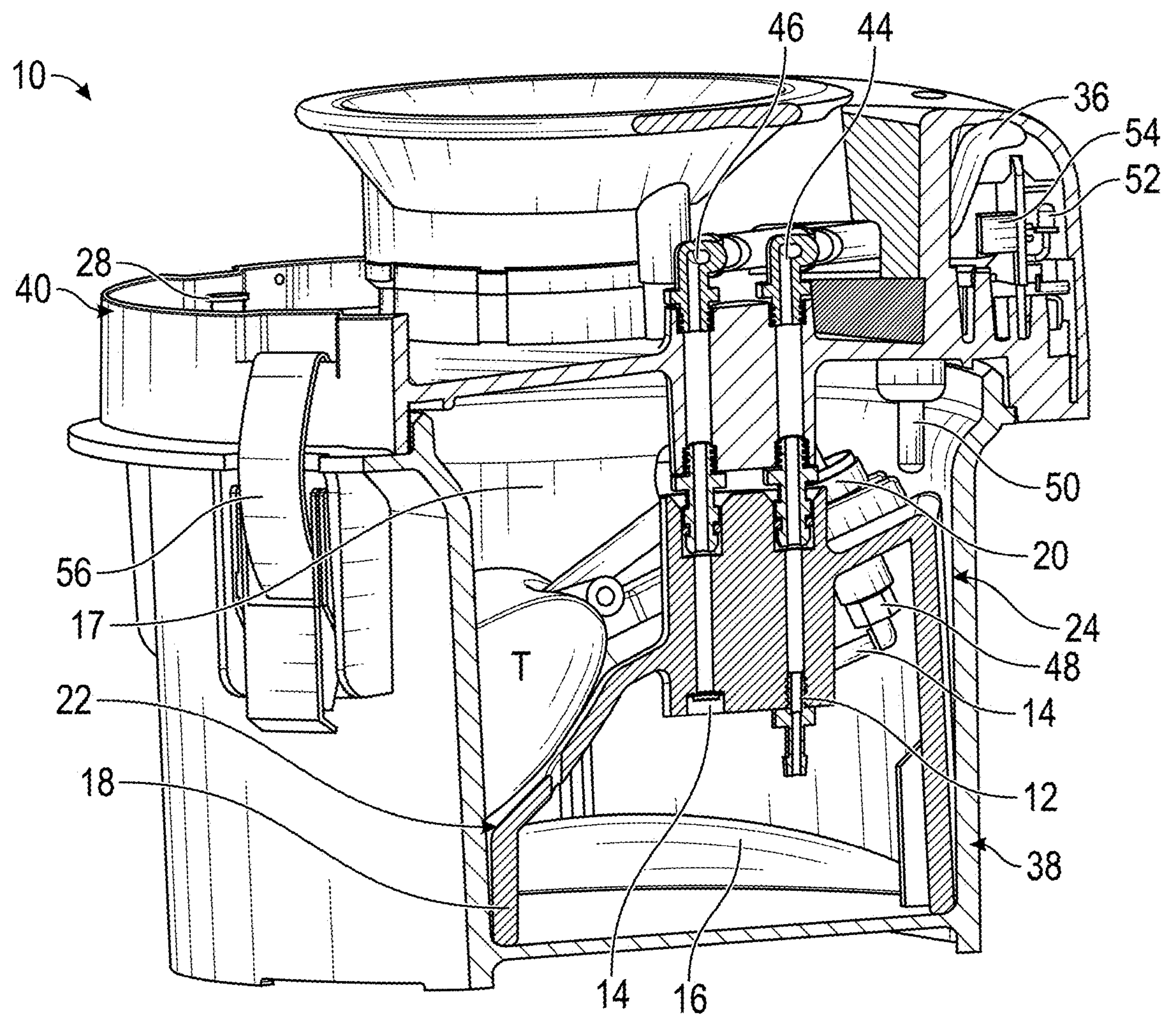


FIG. 1A

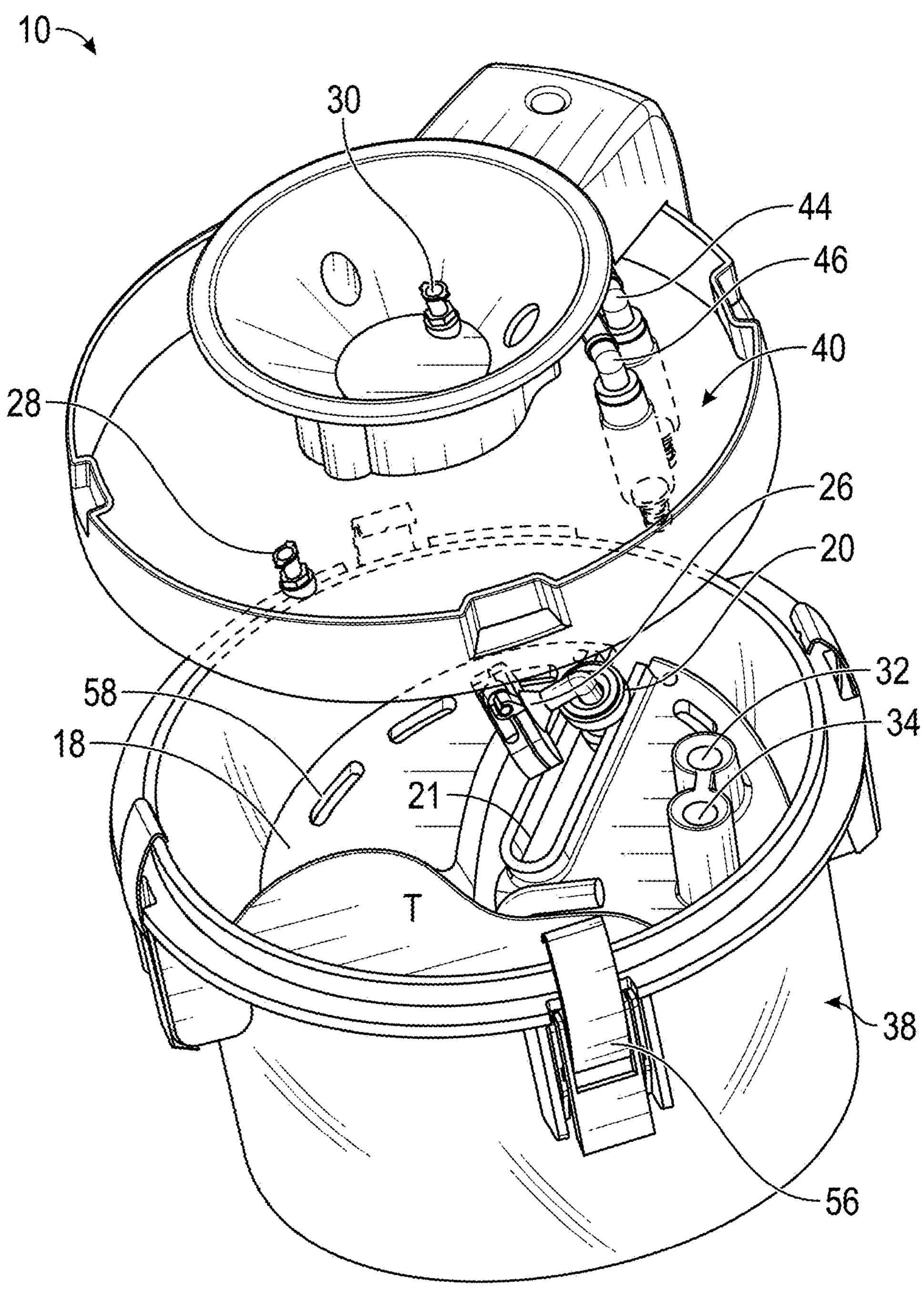


FIG. 1B

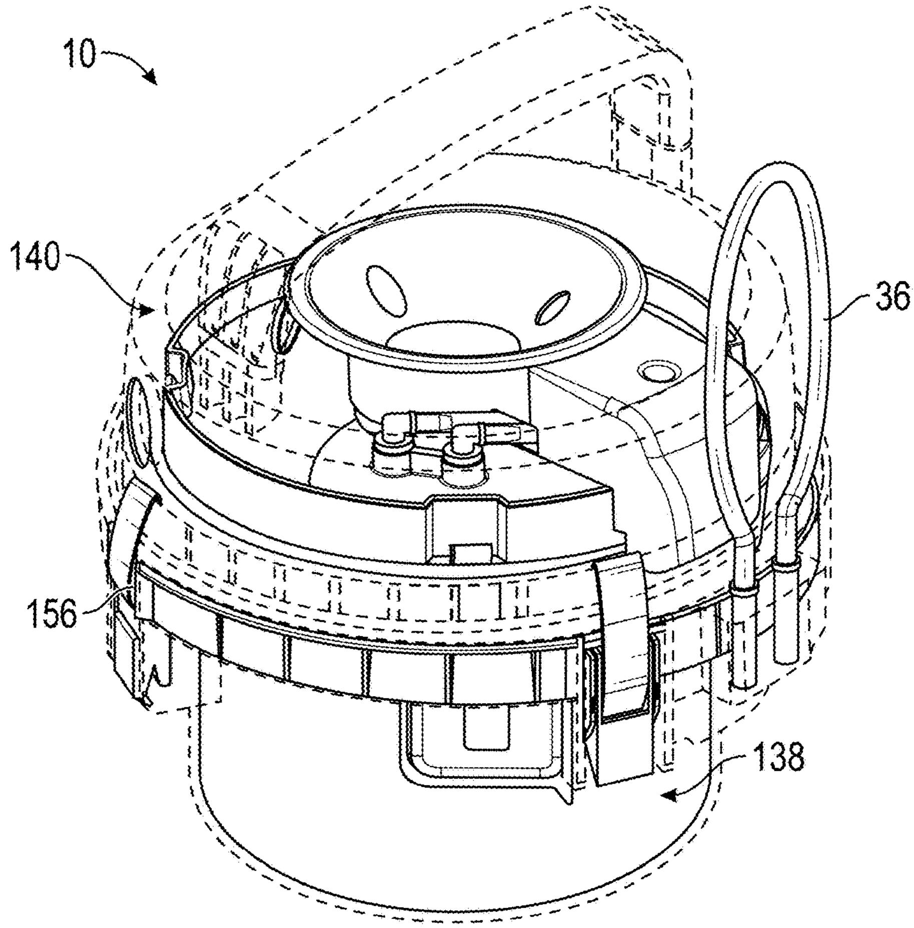


FIG. 1C

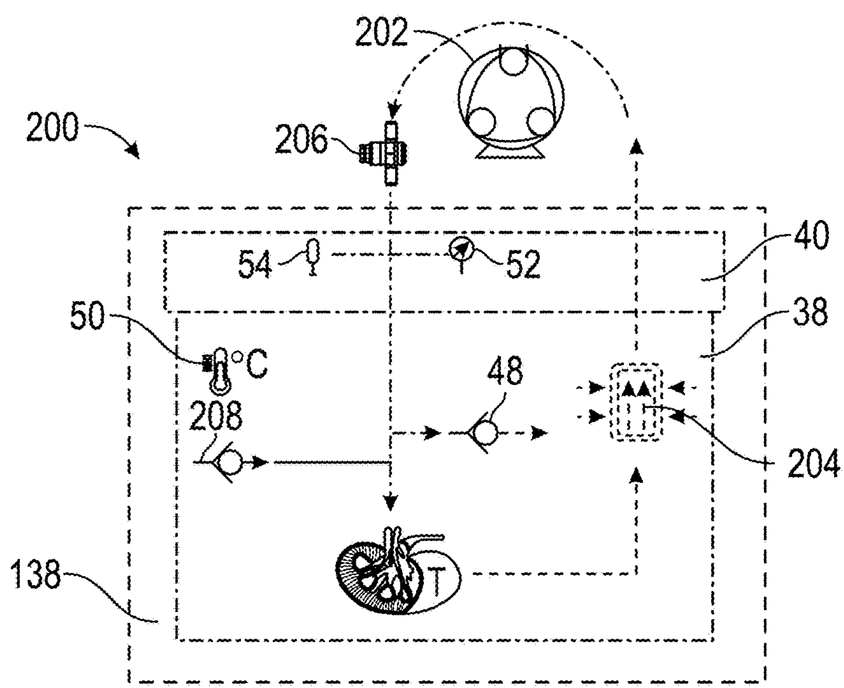


FIG. 2A

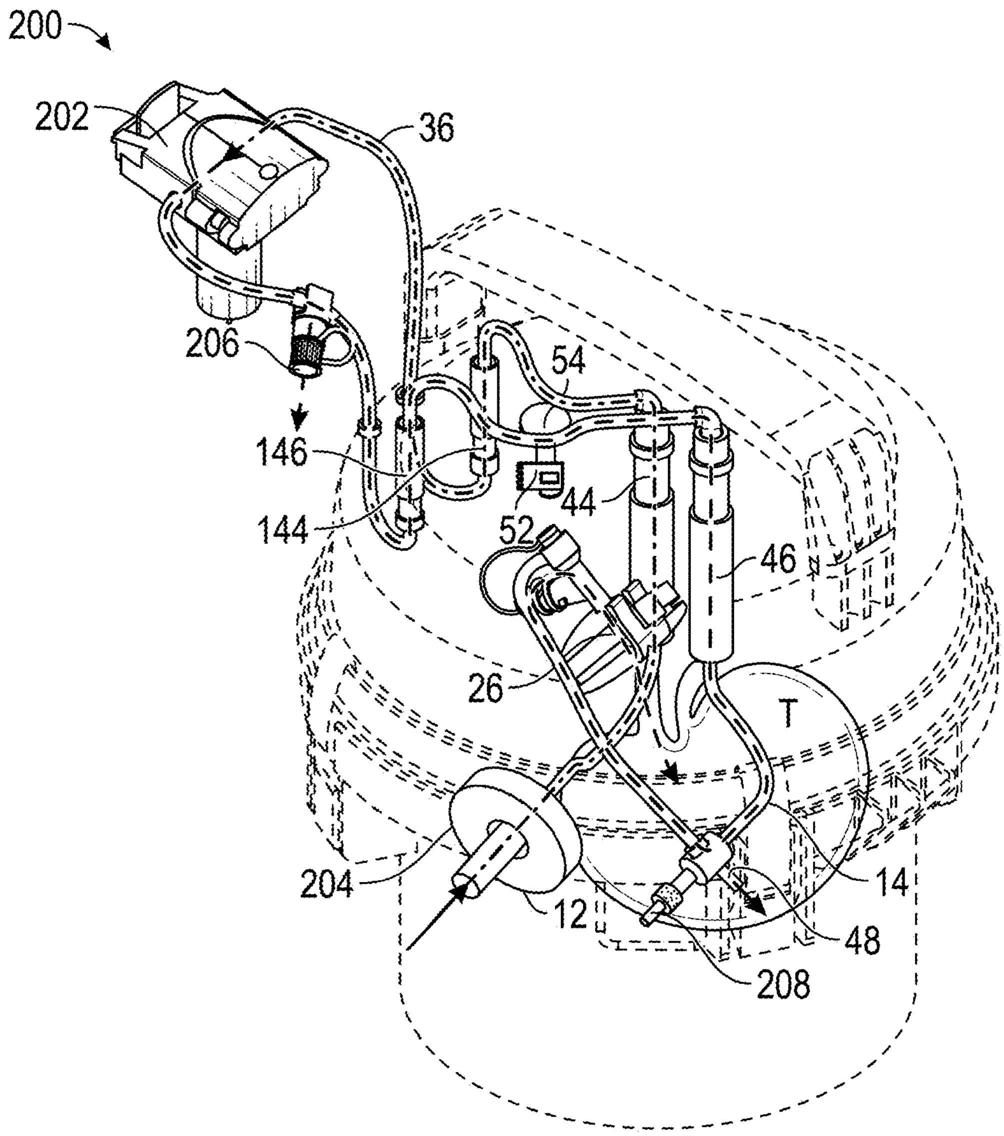


FIG. 2B

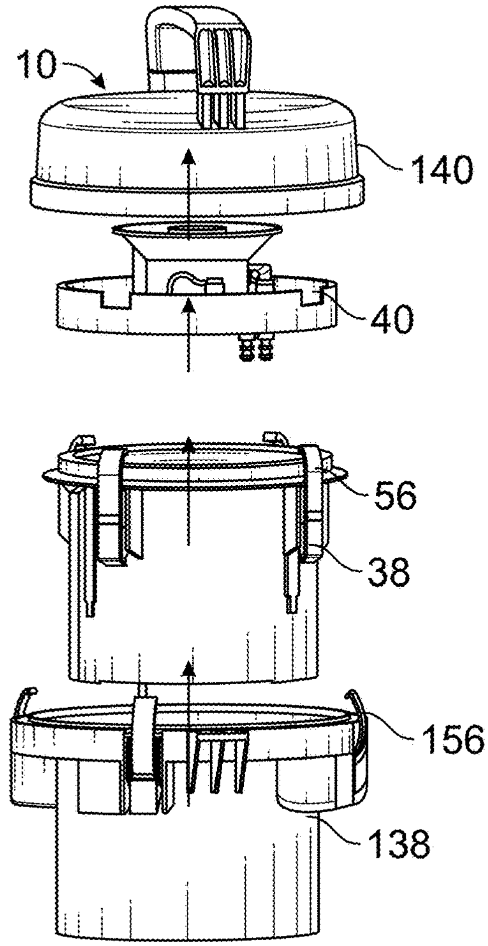


FIG. 3A

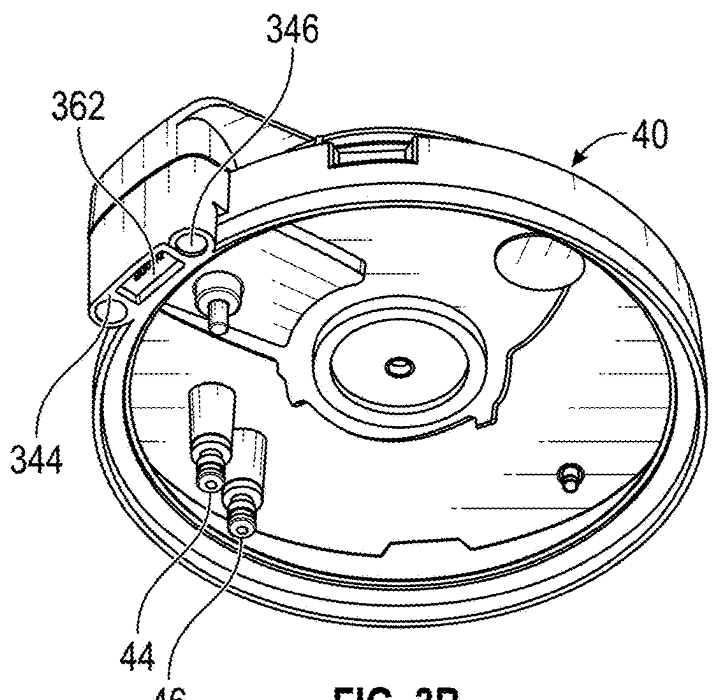


FIG. 3B

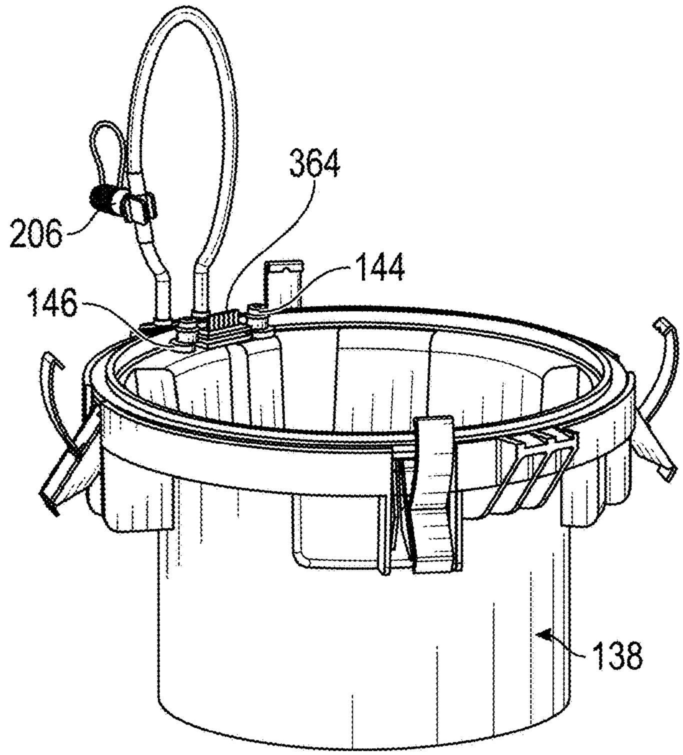


FIG. 3C

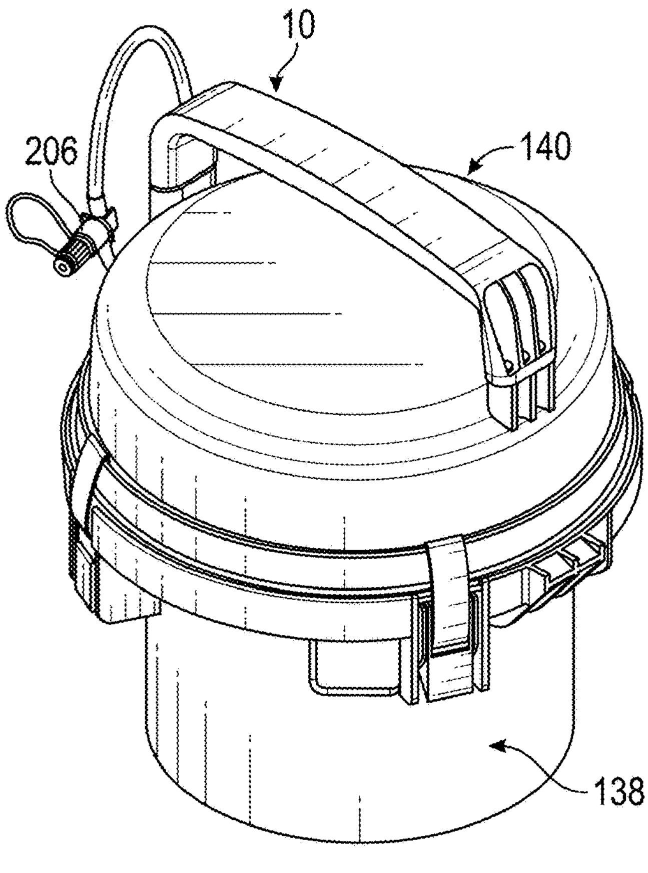
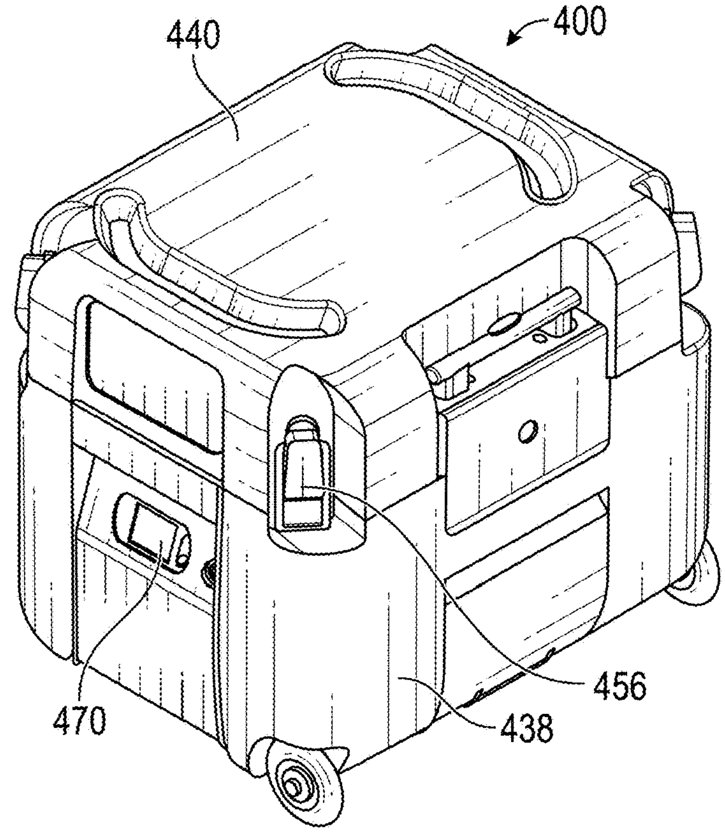


FIG. 3D



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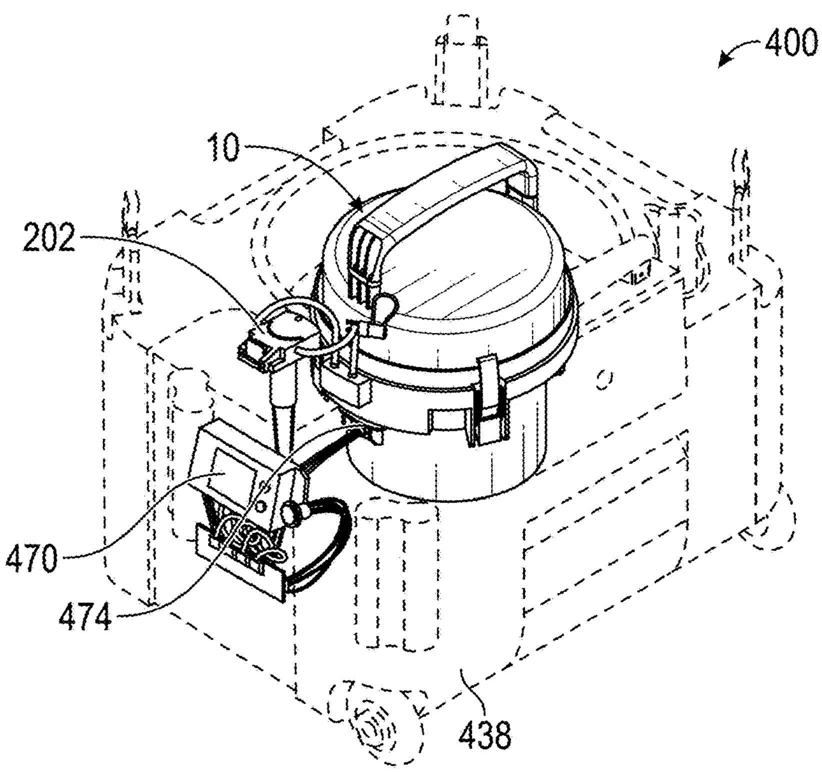


FIG. 4B

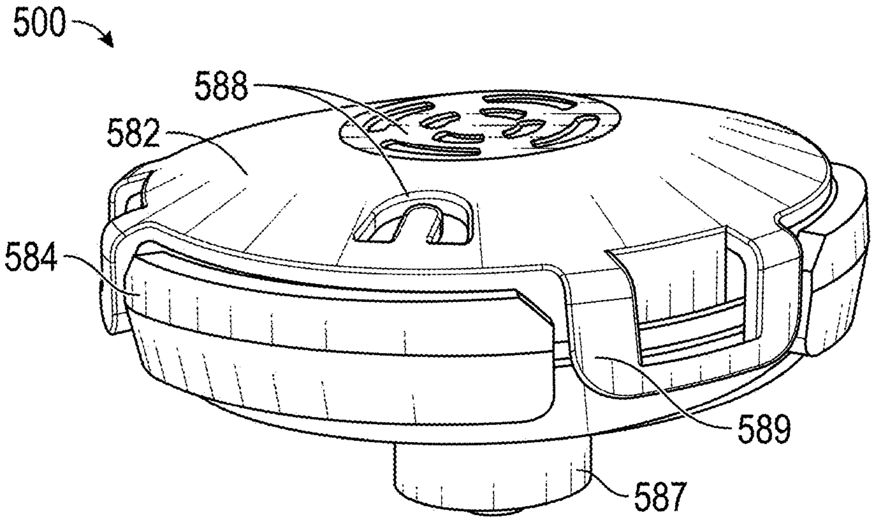


FIG. 5A

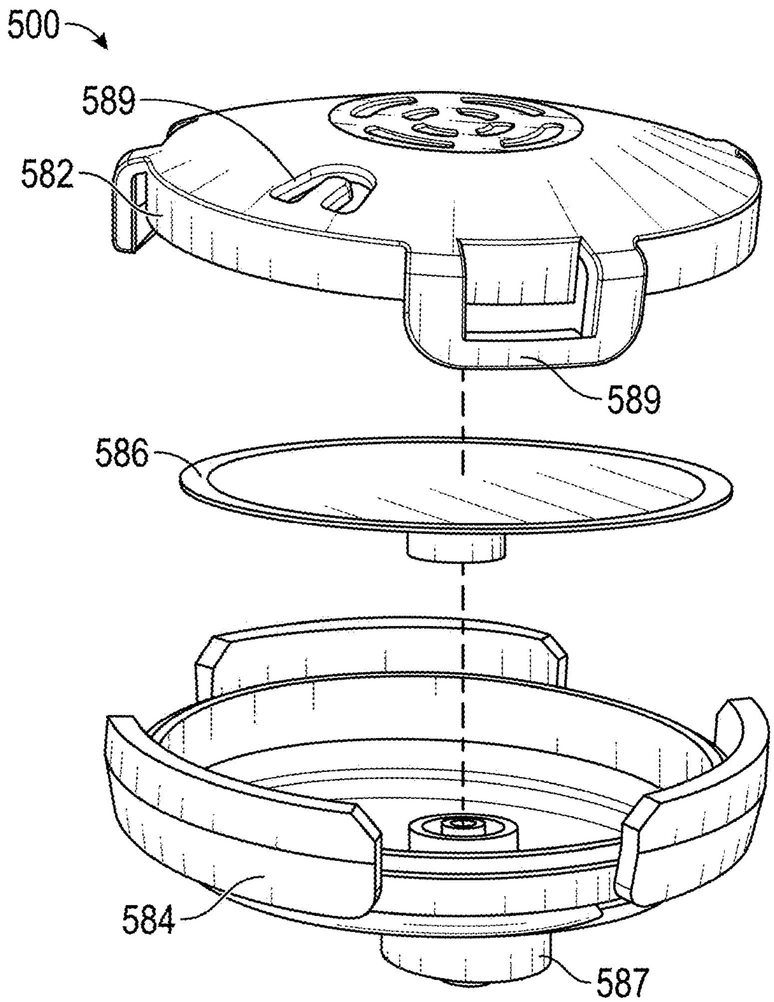


FIG. 5B

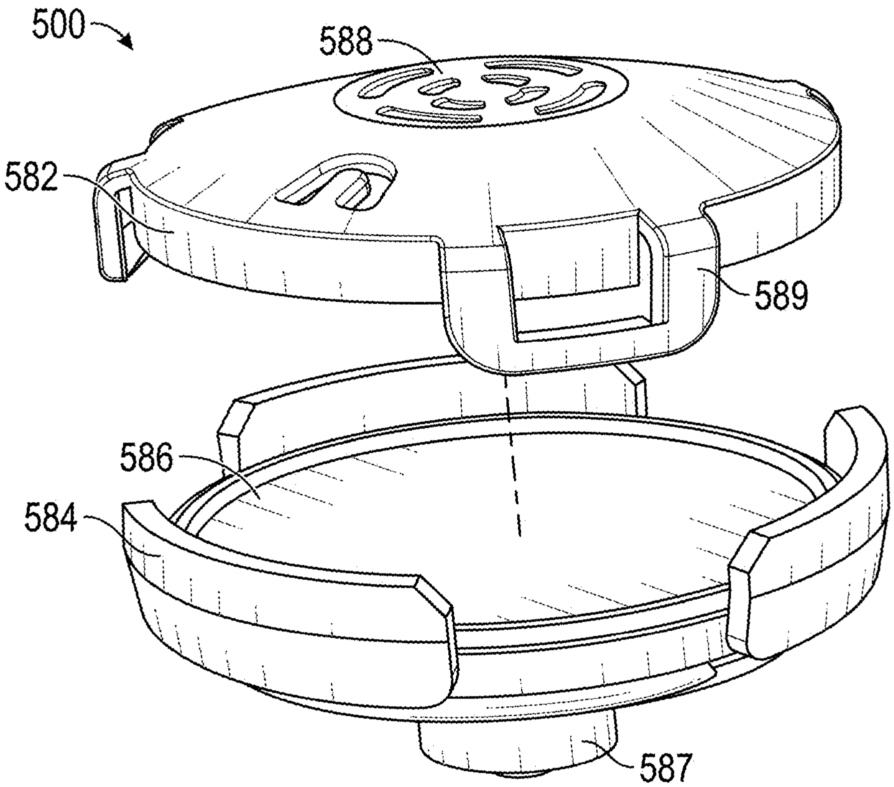


FIG. 5C

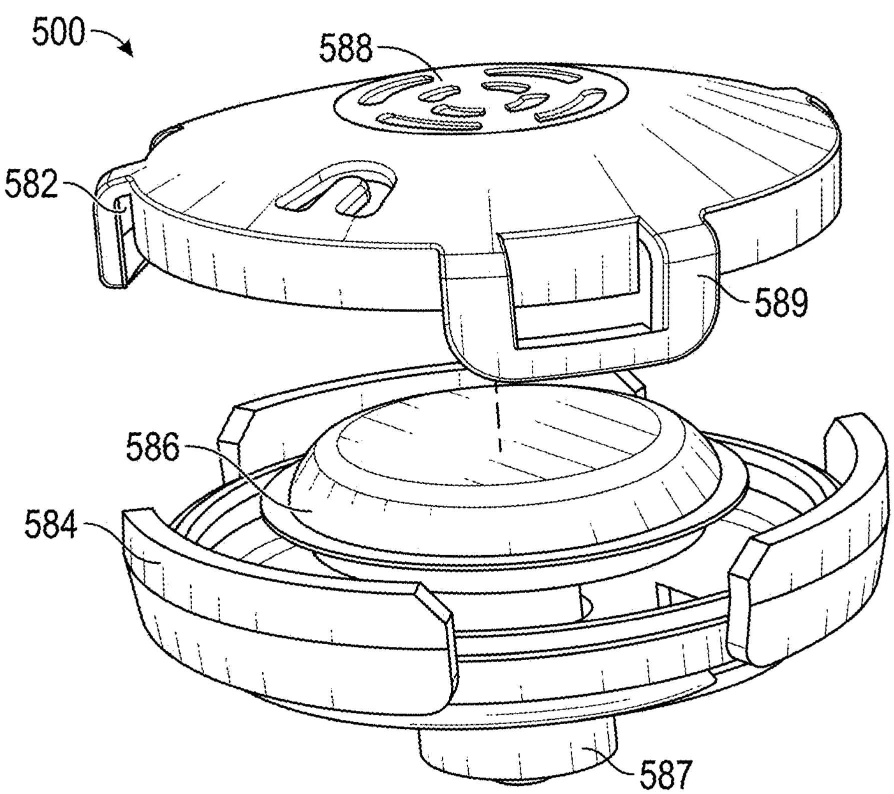


FIG. 5D

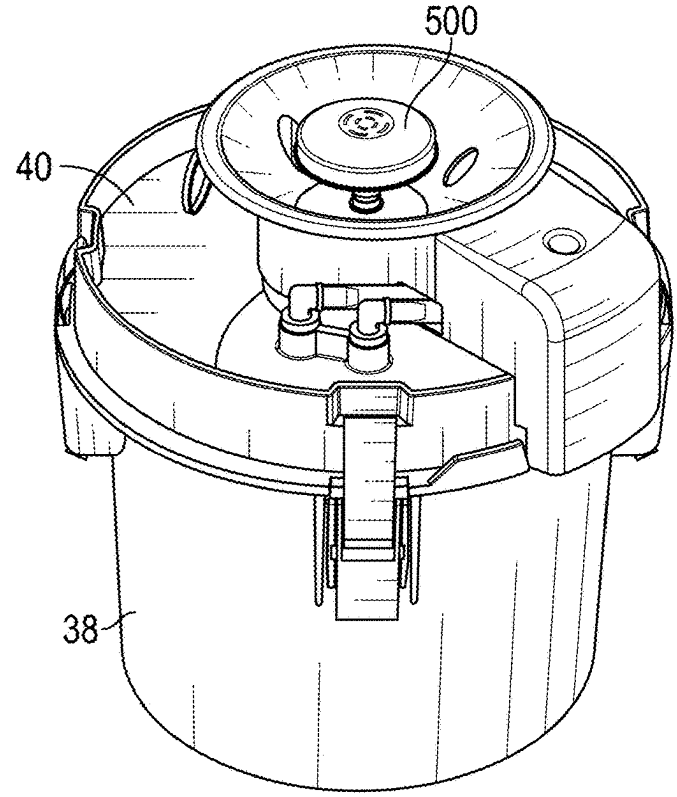


FIG. 6A

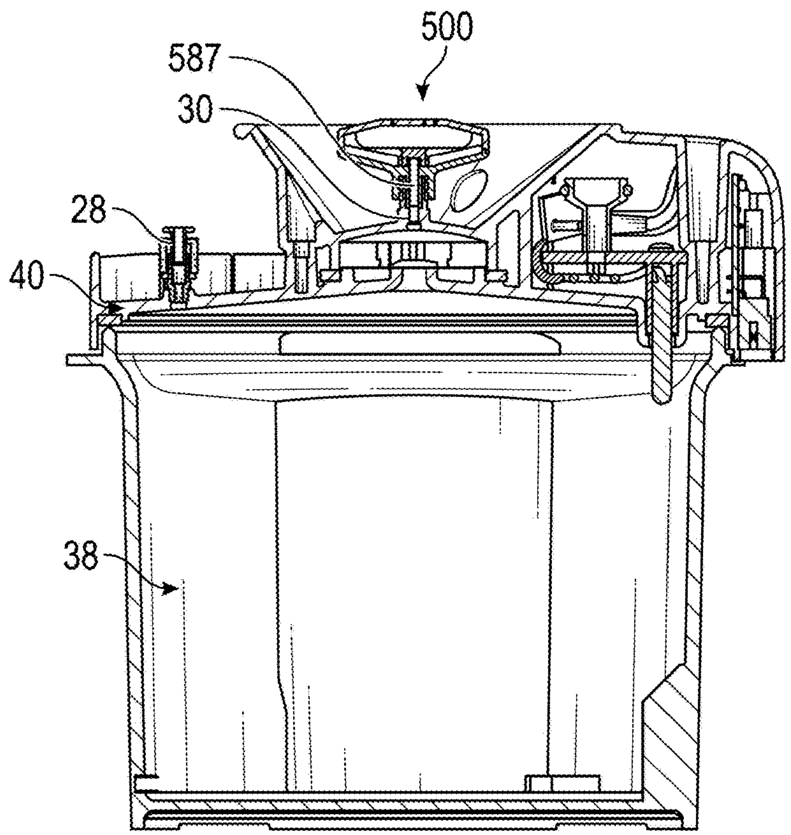


FIG. 6B

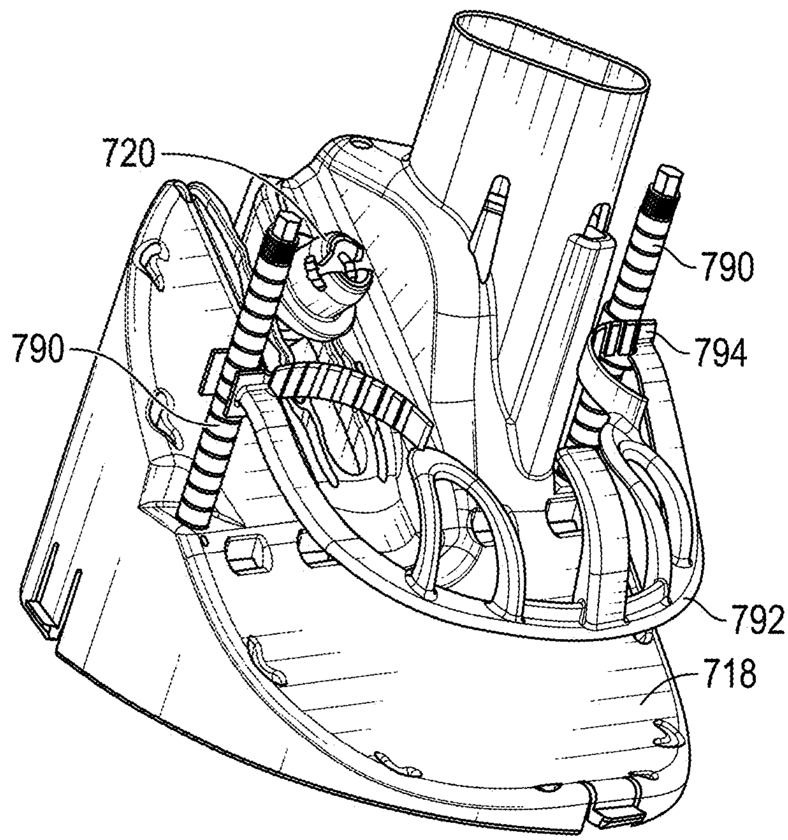


FIG. 7A

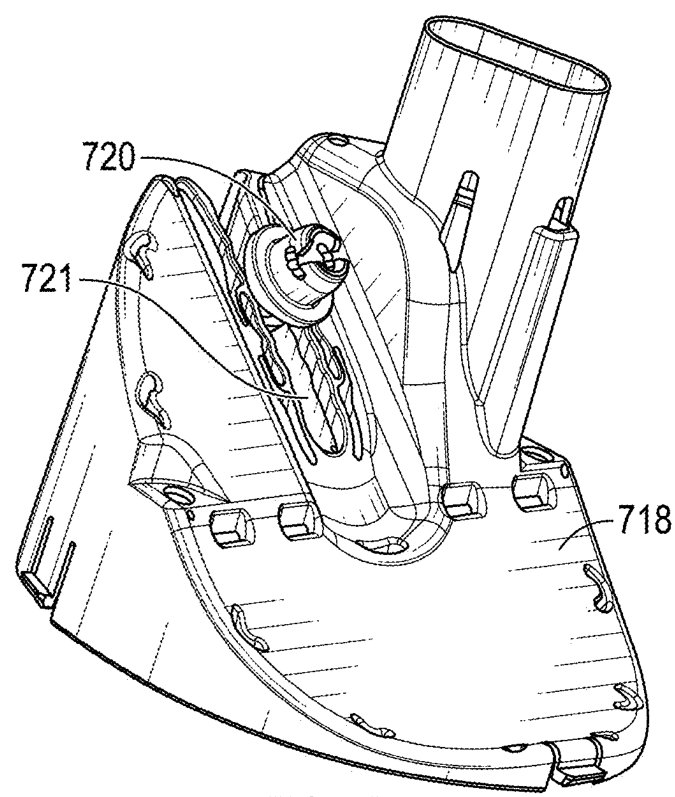


FIG. 7B

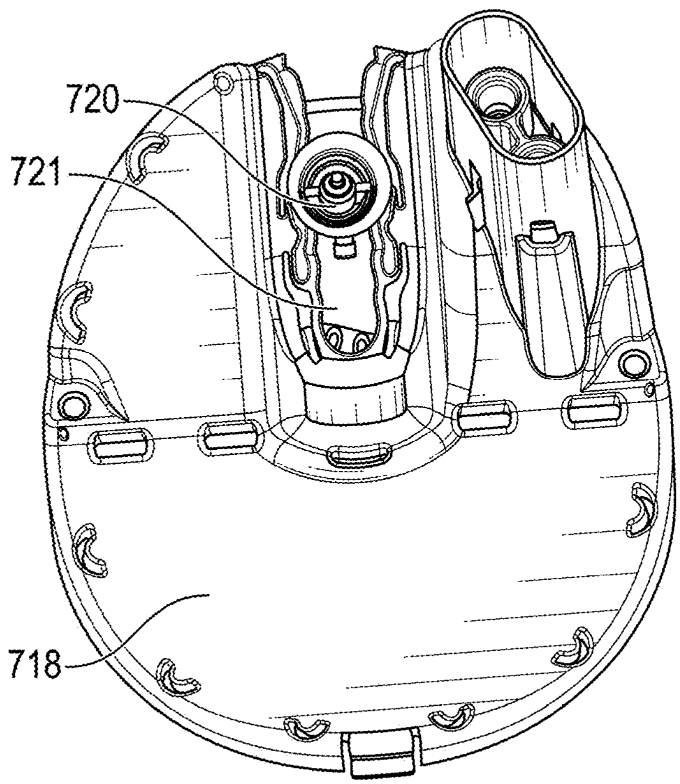


FIG. 7C

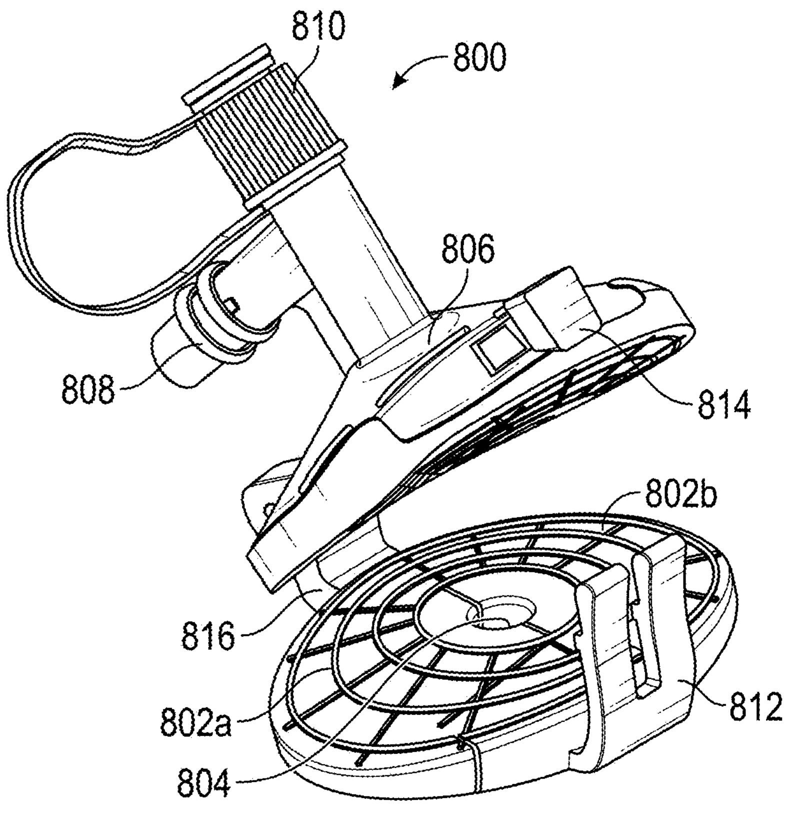


FIG. 8A

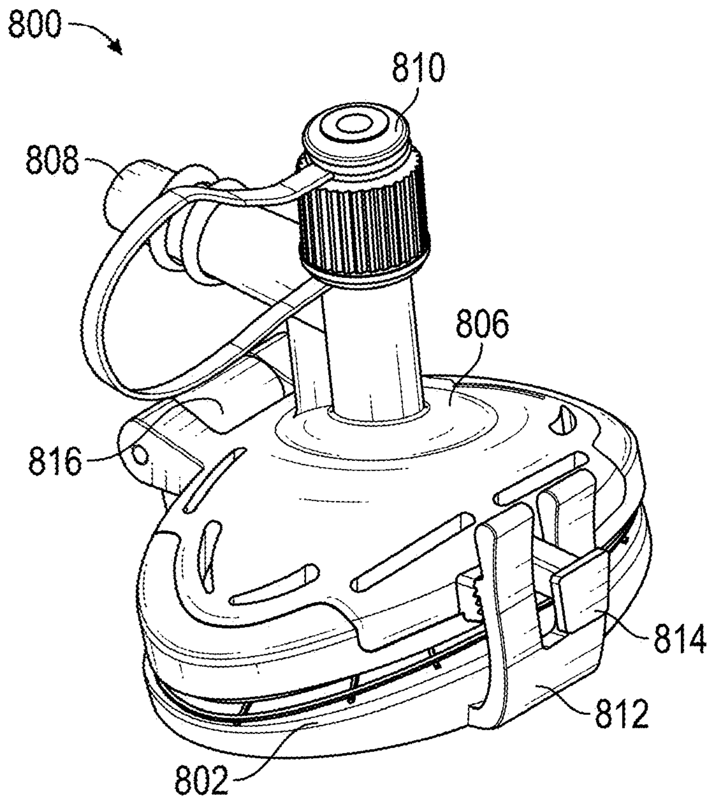

800
810
808
806
816
814
812
802

FIG. 8B

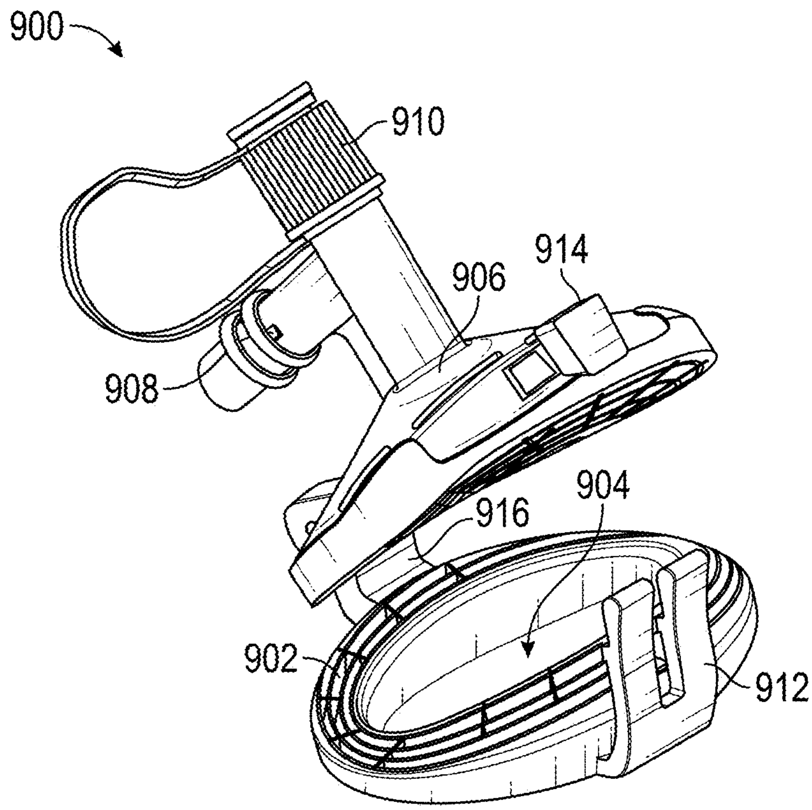

900
910
914
906
908
904
916
902
912

FIG. 9A

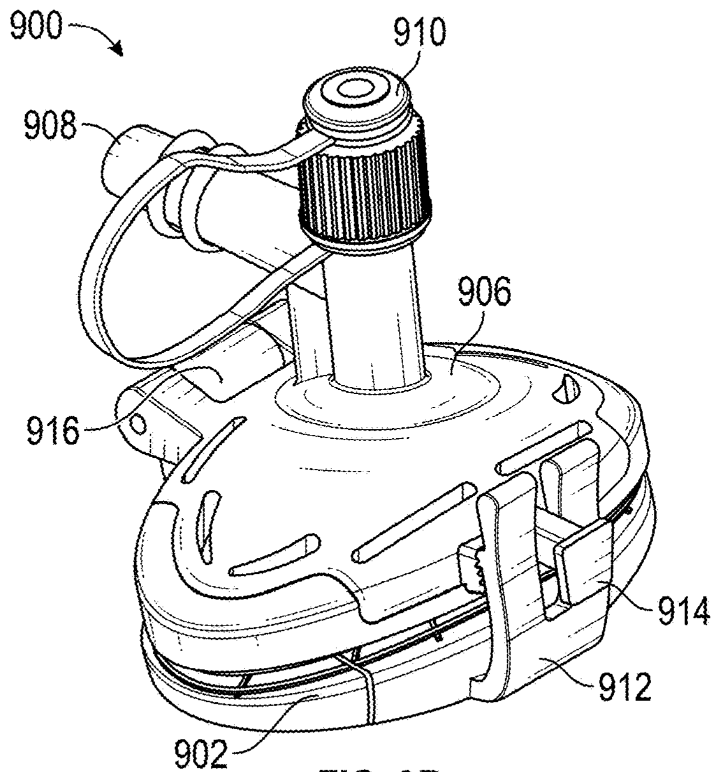


FIG. 9B

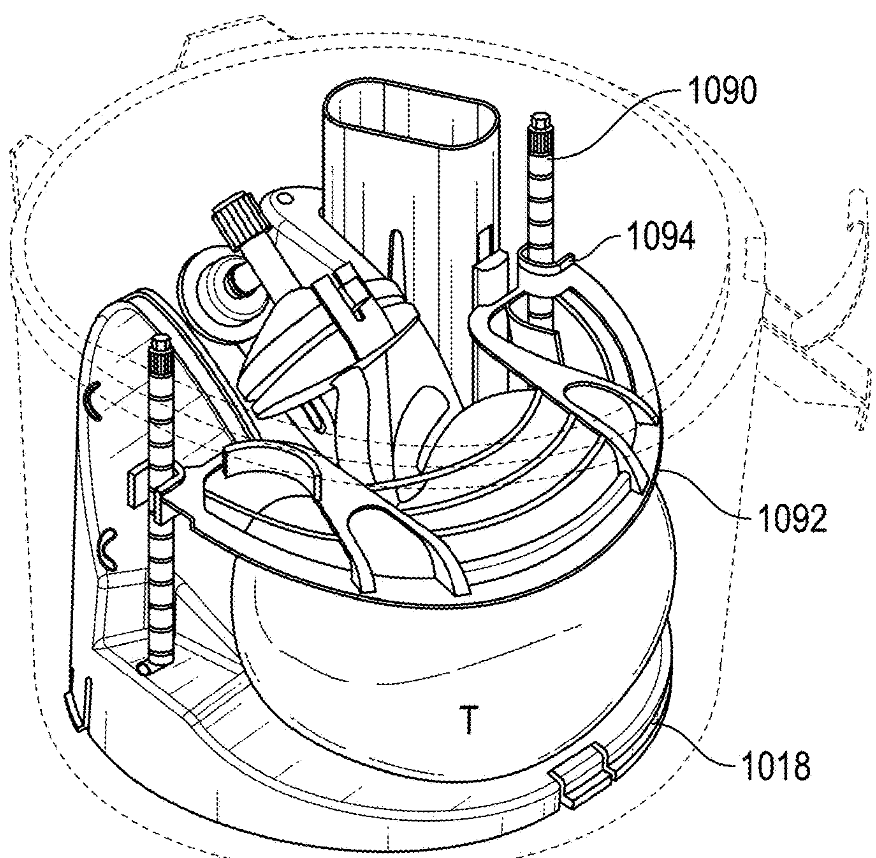
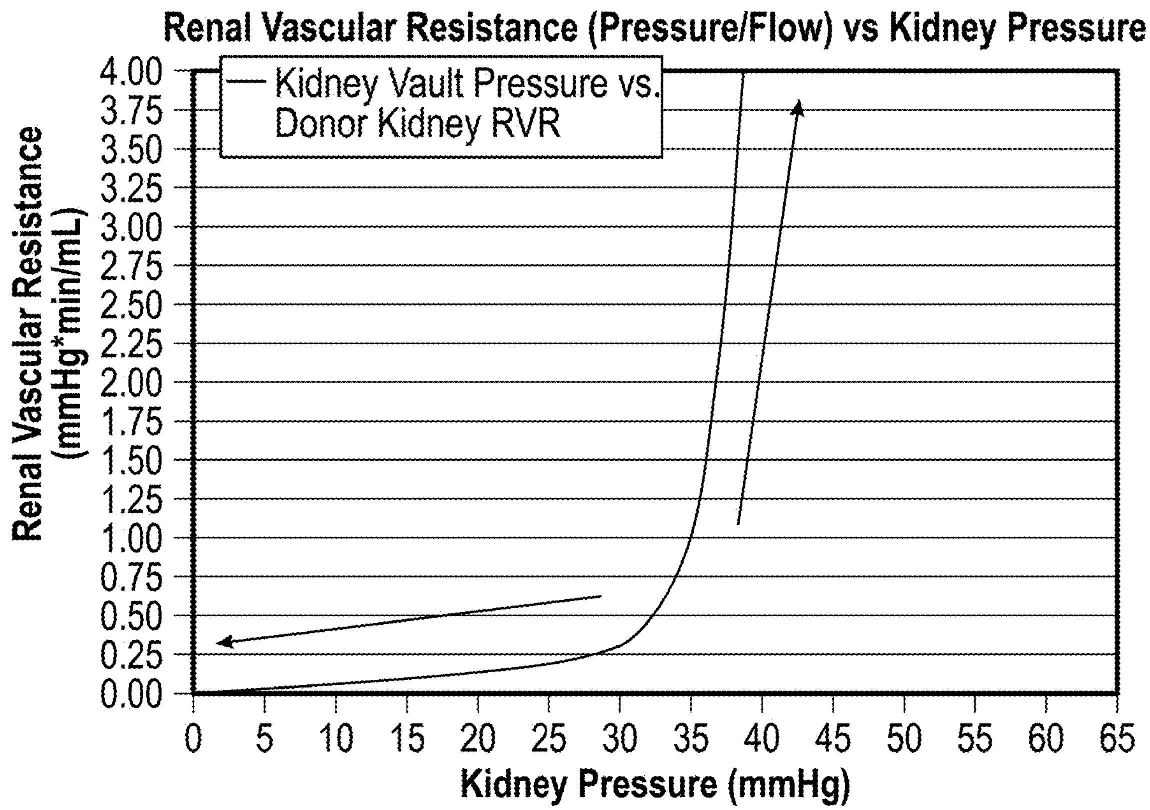
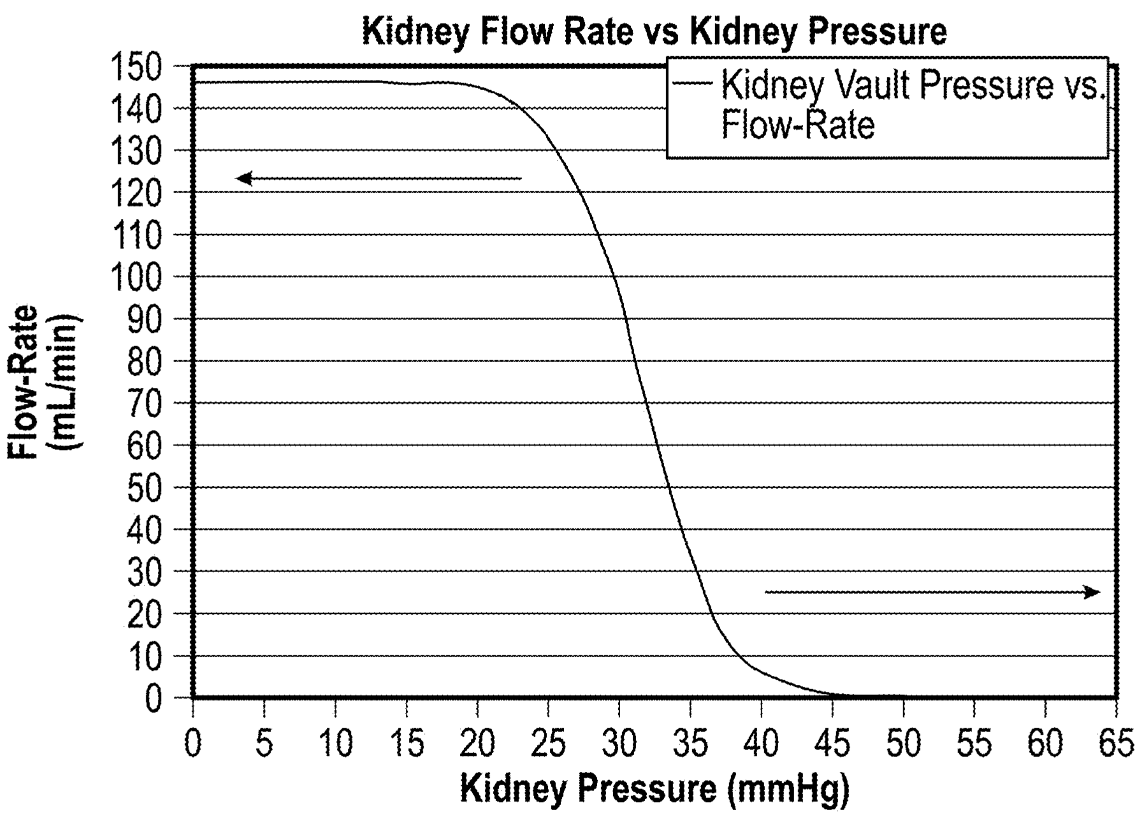


FIG. 10



Renal Vascular Resistance (Pressure/Flow) vs Kidney Pressure
Kidney Vault Pressure vs. Donor Kidney RVR
Renal Vascular Resistance (mmHg*min/mL)
4.00
3.75
3.50
3.25
3.00
2.75
2.50
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0 5 10 15 20 25 30 35 40 45 50 55 60 65
Kidney Pressure (mmHg)

FIG. 11



Kidney Flow Rate vs Kidney Pressure
Kidney Vault Pressure vs. Flow-Rate
Flow-Rate (mL/min)
150
140
130
120
110
100
90
80
70
60
50
40
30
20
10
0
0 5 10 15 20 25 30 35 40 45 50 55 60 65
Kidney Pressure (mmHg)

FIG. 12

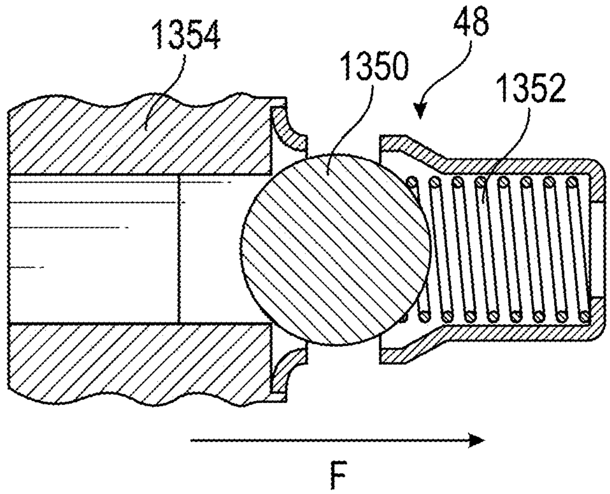


FIG. 13A

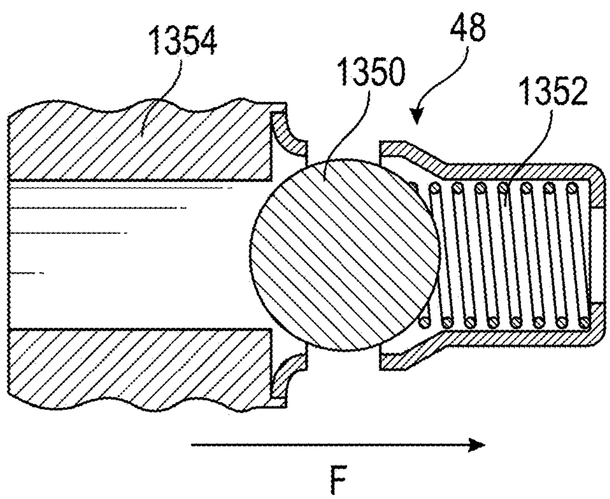


FIG. 13B

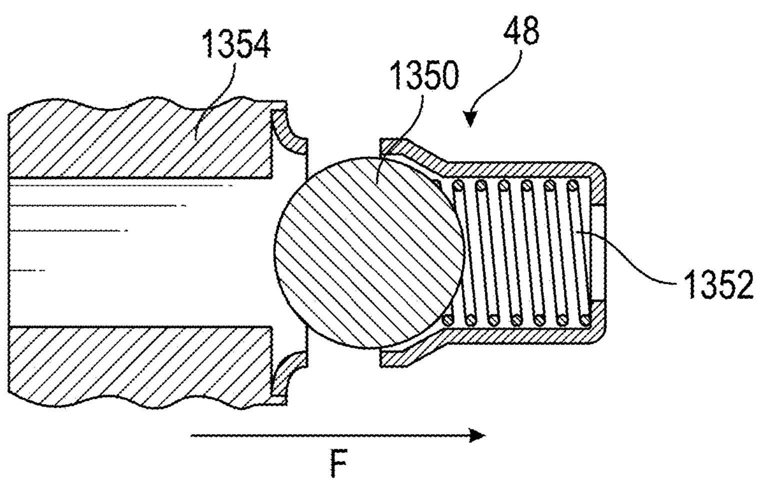


FIG. 13C

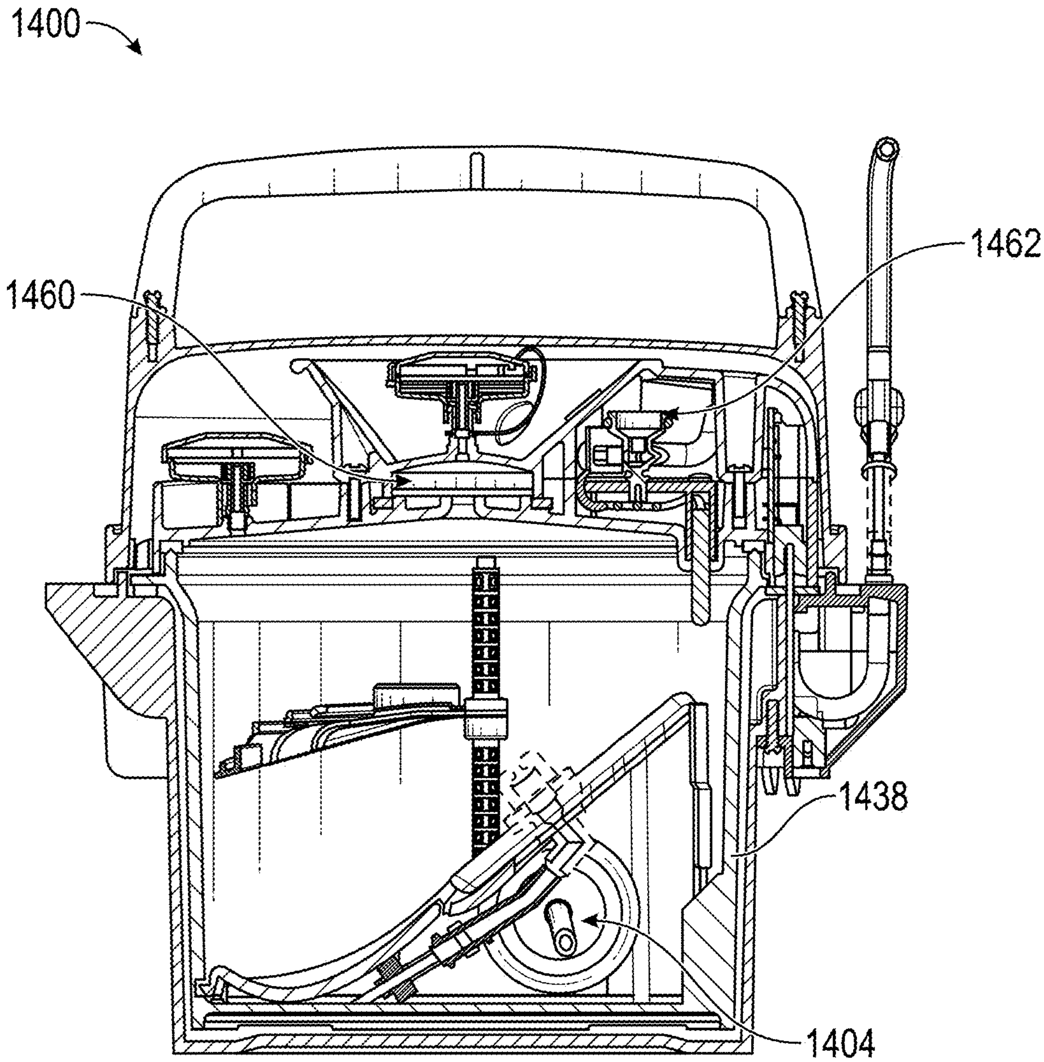


FIG. 14

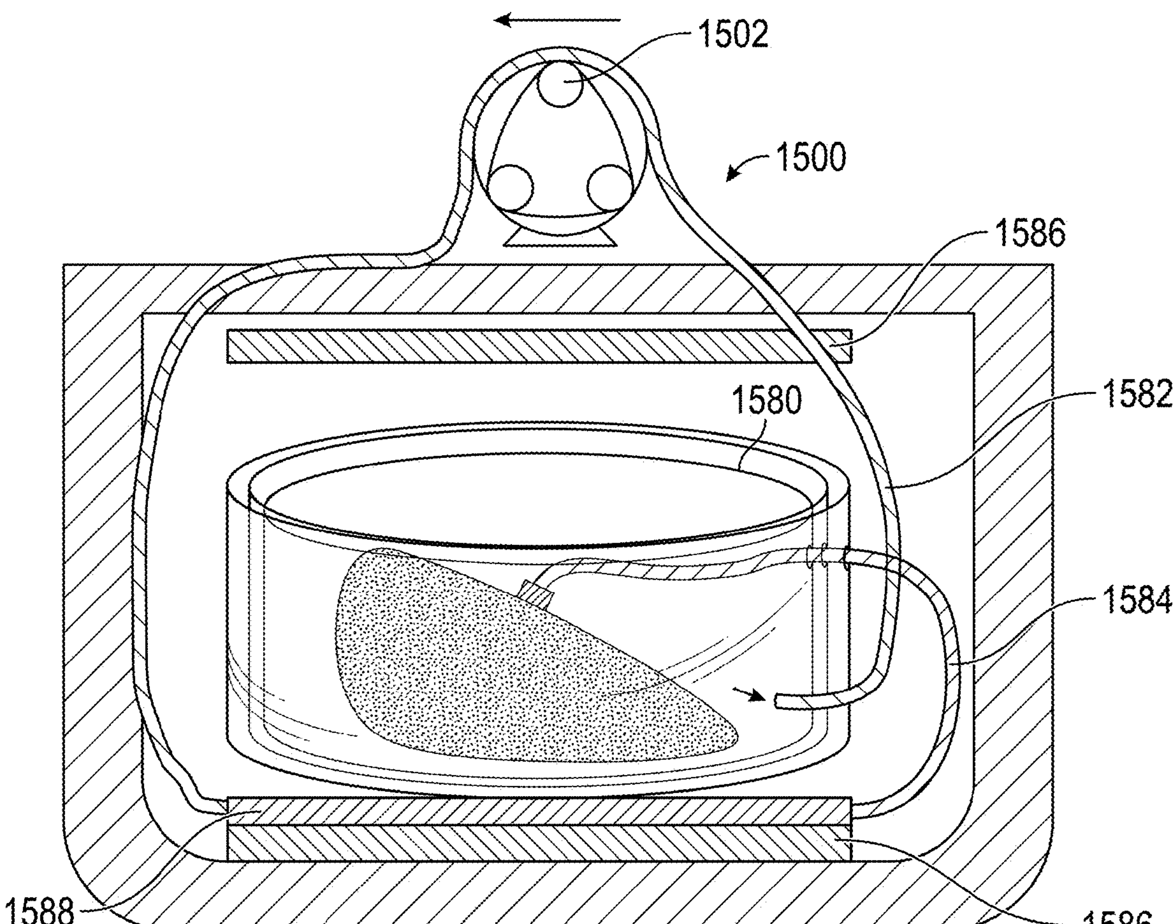


FIG. 15

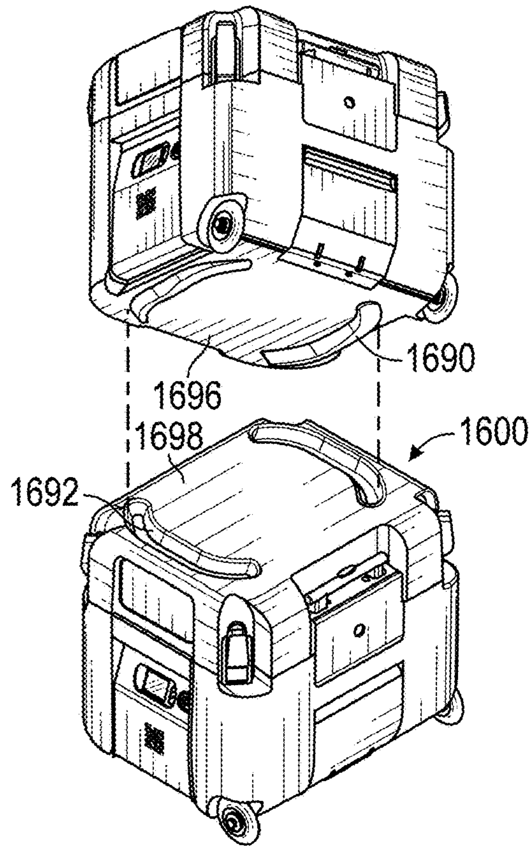


FIG. 16

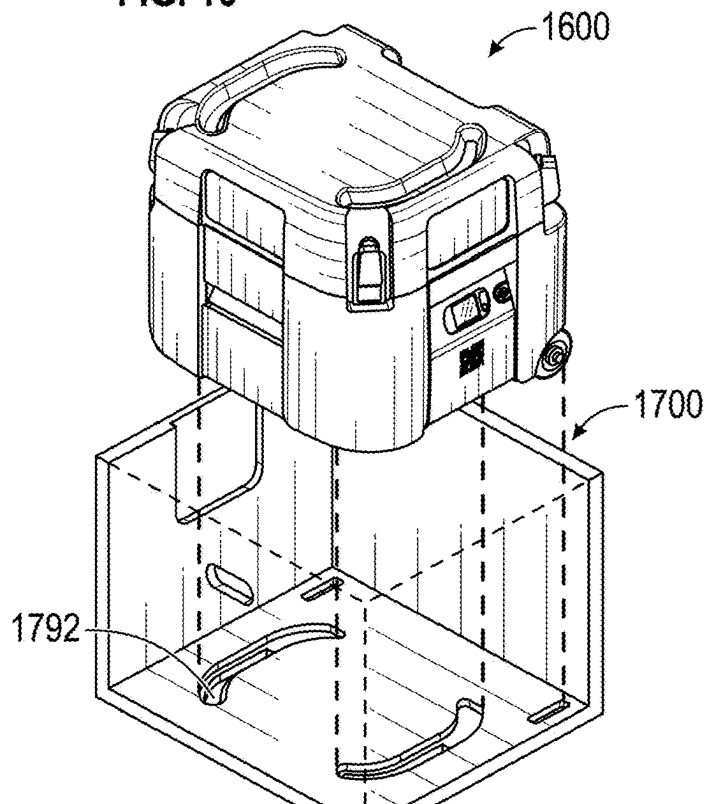


FIG. 17

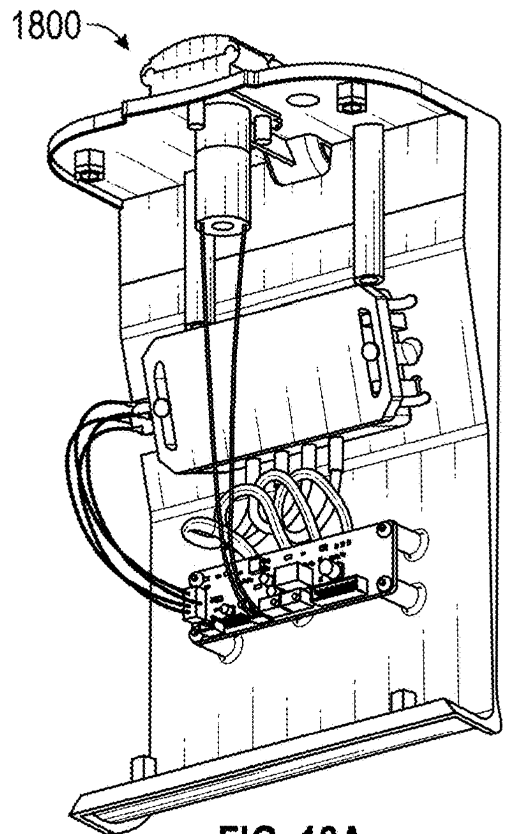


FIG. 18A

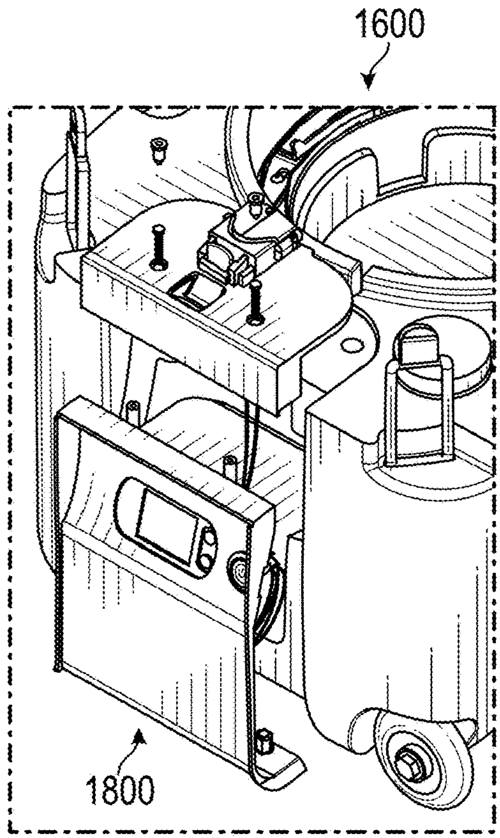


FIG. 18B

METHOD FOR HYPOTHERMIC TRANSPORT OF BIOLOGICAL SAMPLES

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

This application claims the benefit of priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application No. 63/549,145, filed Feb. 2, 2024, and U.S. Provisional Patent Application No. 63/727,110, filed Dec. 2, 2024. Both of these applications are hereby incorporated by reference herein in their entireties. Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57.

FIELD

This disclosure relates to systems and method for hypothermic transport of biological samples, for example tissues for donation. The systems and methods provide a secure, sterile, and temperature-controlled environment for transporting the samples.

BACKGROUND

There is a critical shortage of donor organs. Hundreds of lives could be saved each day if more organs (heart, kidney, lung, etc.) were available for transplant. While the shortage is partly due to a lack of donors, there is a need for better methods of preserving and transporting donated organs. Current storage and preservation methods allow only a small time window between harvest and transplant, typically on the order of hours. These time windows dictate who is eligible to donate organs and who is eligible to receive the donated organs. These time windows also result in eligible organs going unused because they cannot be transported to a recipient in time.

Current organ perfusion devices often require a variable pump to adjust flow rate during transport. This can lead to imprecision due to lag time and/or human error. Certain current organ perfusion devices draw fluid from an external source rather than using a closed loop system, which can result in complications due to additional variables.

Current organ perfusion devices typically utilize an expensive, re-usable capital component (pump) and single-use components (cassette), and require user input and monitoring for temperature, perfusion pressure, and renal flow. The need for active monitoring by a health care provider who must listen for alarms and intervene (e.g., add ice to ensure adequate kidney cooling) and adjust settings during transport create staffing challenges and limits the options for transporting organs to the transplant recipients. Specifically, it has become common for kidneys, unlike other organs, to be transported by air unaccompanied. In such, it is impractical to transport kidneys in current perfusion devices unaccompanied.

Improved transport and storage for organs would increase the pool of available organs while improving outcomes for recipients.

SUMMARY

The disclosure provides an improved system for transporting biological samples, e.g., tissues, such as donor organs. In certain examples, this improved system may greatly expand the window of time for organ transportation and, consequently, make many more organs available for donation. Additionally, the samples may be healthier upon arrival, as compared to state-of-the-art transport methods. In some embodiments, organ perfusion can prolong viability of donor organs, for example kidneys, by ensuring uniform cooling and flushing out of metabolites. Current organ perfusion devices often require monitoring during transportation to ensure that the fluid flow is within a desired range. Such an approach can require a user to increase or decrease a pump speed, either manually or by an automatic controller. In systems that use an automatic controller, the pump parameters can be determined by an algorithm based on pressure sensor measurements. Such as approach may lead to imprecision based on inaccuracy of the sensors and the algorithm. Further, the feedback loop can also cause delay in adjusting the pump parameters.

Embodiments of the disclosed system for organ transport overcome the shortcomings of the prior art by delivering a flow of preservation fluid at pressures acceptable to the organ anatomy without requiring manual operation or a control algorithm. For example, the flow regulator valve can ensure that flow through the organ is acceptable, varying the amount of fluid pumped into the vessel based on the vessel's resistance. The flow regulator valve can provide a mechanical check that is precise and instantaneous. A pressure dampener can be used to provide constant flow, reducing pulsation caused by the peristaltic pump. The flow regulator valve in combination with the pressure dampener can be especially advantageous as it ensures constant flow through the organ perfusion system.

In some embodiments, the systems described herein relate to hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, including: a transport container configured to receive an organ, the transport container configured to be at least partially filled with preservation fluid, the transport container including: a fluid circuit including: an inlet configured to be in communication with the preservation fluid in the transport container; a flow regulator valve in fluid communication with the inlet, the flow regulator valve configured to release the preservation fluid from the fluid circuit into the transport container at a rate based on renal resistance of the organ; and an outlet in fluid communication with the flow regulator valve and the organ; and a pump configured to pump the preservation fluid from the inlet to the organ, the pump configured to operate independently of renal resistance.

In some embodiments, the transport container further includes an organ rest configured to support the organ. In some embodiments, the systems described herein include an organ adapter configured to seal a vessel of the organ, the organ adapter adjustable along a groove of the organ rest. In some embodiments, the systems described herein include a pressure dampener, wherein the pressure dampener is configured to reduce pulsation in the fluid circuit caused by the pump. In some embodiments, the pump is a peristaltic pump. In some embodiments, the pump is a centrifugal pump, a diaphragm pump, a piston pump, or a gear pump. In some embodiments, the flow regulator valve releases the preservation fluid from the fluid circuit into the transport container at a higher rate when renal resistance is higher. In some embodiments, the transport container further includes: a plurality of posts; and an organ retainer adjustably couplable to the plurality of posts. In some embodiments, the transport container further includes a one-way valve in fluid communication with the fluid circuit, the one-way valve configured to take in preservation fluid from the transport container when fluid is drawn out of the fluid circuit. In some embodiments, the transport container further includes a temperature sensor. In some embodiments, the systems described herein include a lid including a fill port and a vent port. In some embodiments, the systems described herein include an accumulation chamber configured to seal to at least one of the fill port and the vent port, the accumulation chamber including a balloon configured to expand when the preservation fluid expands and contract when the preservation fluid contracts. In some embodiments, the organ adapter includes: a cannula including a barb, the cannula configured to seal to the vessel of the organ; and a cannula receiver configured to couple with the barb.

In some embodiments, the methods described herein relate to hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, the method including: placing an organ in a transport container, the transport container including an organ adapter; sealing the organ adapter to a vessel of the organ; placing a lid on the transport container; filling the transport container at least partially with preservation fluid through a fill port in the lid; and activating a pump, the pump configured to pump the preservation fluid from the transport container to a flow regulator valve and the organ adapter, the flow regulator valve configured to release the preservation fluid into the transport container at a rate based on renal resistance of the organ, and the pump configured to operate independently of renal resistance.

In some embodiments, the methods described herein include placing the organ on an organ rest. In some embodiments, the methods described herein include adjusting the organ adapter along a groove of the organ rest. In some embodiments, the methods described herein include attaching an organ retainer to a plurality posts in the transport container such that the organ retainer secures the organ. In some embodiments, the methods described herein include sealing an accumulation chamber to at least one of the fill port or a vent port on the lid.

In some embodiments, the systems described herein relate to hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, including: a transport container configured to receive an organ, the transport container configured to be at least partially filled with preservation fluid; a pump configured to pump the preservation fluid from the transport container to the organ in a fluid circuit, the pump configured to operate independently of renal resistance; and a flow regulator valve in fluid communication with the fluid circuit, the flow regulator valve configured to release the preservation fluid from the fluid circuit into the transport container at a rate based on renal resistance of the organ.

In some embodiments, the transport container further includes an organ rest configured to support the organ. In some embodiments, the systems described herein include an organ adapter configured to seal a vessel of the organ, the organ adapter adjustable along a groove of the organ rest. In some embodiments, the systems described herein include a pressure dampener, wherein the pressure dampener is configured to reduce pulsation caused by the pump. In some embodiments, the pump is a peristaltic pump. In some embodiments, the pump is a centrifugal pump, a diaphragm pump, a piston pump, or a gear pump. In some embodiments, the flow regulator valve releases the preservation fluid from the fluid circuit into the transport container at a higher rate when renal resistance is higher. In some embodiments, the transport container further includes: a plurality of posts; and an organ retainer adjustably couplable to the plurality of posts. In some embodiments, the transport container further includes a one-way valve in fluid communication with the fluid circuit, the one-way valve configured to take in preservation fluid from the transport container when fluid is drawn out of the fluid circuit. In some embodiments, the transport container further includes a temperature sensor. In some embodiments, the systems described herein include a lid including a fill port and a vent port. In some embodiments, the systems described herein include an accumulation chamber configured to seal to at least one of the fill port and the vent port, the accumulation chamber including a balloon configured to expand when the preservation fluid expands and contract when the preservation fluid contracts. In some embodiments, the organ adapter includes: a cannula including a barb, the cannula configured to seal to the vessel of the organ; and a cannula receiver configured to couple with the barb.

In some embodiments, the systems described herein relate to hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, including: a canister configured to contain an organ and preservation fluid; an inner lid configured to couple with the canister; an outer canister configured to couple with the inner lid, the outer canister including a tube; and a pump configured to pump the preservation fluid in the tube such that the preservation fluid flows from the canister to the inner lid, from the inner lid to the outer canister, from the outer canister to the tube, from the tube to the outer canister, from the outer canister to the inner lid, from the inner lid to the organ, and from the organ to the canister.

In some embodiments, the systems described herein relate to hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, including: a canister configured to receive an organ, the canister containing preservation fluid; a port between the canister and an environment; and an accumulation chamber configured to seal to the port, the accumulation chamber including a balloon configured to expand when the preservation fluid expands and contract when the preservation fluid contracts.

In some embodiments, the methods described herein relate to hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, the method including: placing an organ in a canister; filling the canister with preservation fluid through a fill port; venting the canister of fluid through a vent port; and attaching an accumulation chamber on at least one of the fill port or the vent port, the accumulation chamber including a balloon configured to expand when the preservation fluid expands and contract when the preservation fluid contracts.

In some embodiments, the systems described herein relate to hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, including: a canister configured to receive an organ, the canister including: an organ rest configured to support the organ; and a plurality of posts; and an organ retainer adjustably couplable to the plurality of posts.

In some embodiments, the methods described herein relate to hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, the method including: placing an organ in a canister, the canister including an organ rest configured to support the organ and a plurality of posts; and coupling an organ retainer to the plurality of posts at a position such that the organ retainer contacts the organ.

In some embodiments, the systems described herein relate to hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, including: a canister configured to receive an organ; an organ rest inside the canister configured to support the organ, the organ rest including a groove; and an organ adapter configured to seal to a vessel of the organ, the organ adapter movable along the groove.

In some embodiments, the organ adapter includes: a cannula configured to seal to the vessel of the organ; and a cannula receiver on the organ rest configured to couple with the cannula.

In some embodiments, the methods described herein relate to hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, the method including: placing an organ in a canister on an organ rest, the organ rest including a groove; sealing an organ adapter to a vessel of the organ; moving the organ adapter along the groove; and locking the organ adapter in place such that a renal artery of the organ is under tension.

In some embodiments, sealing the organ adapter to the vessel of the organ includes: sealing the vessel of the organ with a cannula; and coupling the cannula to a cannula receiver on the organ rest.

In some embodiments, the techniques described herein relate to an apparatus substantially as shown and/or described. In some embodiments, the techniques described herein relate to a method substantially as shown and/or described. In some embodiments, the techniques described herein relate to a system substantially as shown and/or described.

In some examples, the systems for hypothermic transport of a biological sample described herein can include: a transport container configured to receive an organ, the transport container configured to be at least partially filled with preservation fluid, the transport container including: a fluid circuit including: an inlet configured to be in communication with the preservation fluid in the transport container; a flow regulator valve in fluid communication with the inlet, the flow regulator valve configured to regulate a flow rate from the fluid circuit to the organ based on at least one organ parameter; and an outlet in fluid communication with the flow regulator valve and the organ; and a pump configured to pump the preservation fluid from the inlet to the organ, the pump configured to operate independently of the at least one organ parameter.

In some examples, the at least one organ parameter includes at least one of vessel resistance, organ temperature, or flow rate through the organ. In some examples, the transport container further includes an organ rest configured to support the organ. In some examples, the system can include an organ adapter configured to seal a vessel of the organ, the organ adapter adjustable along a groove of the organ rest. In some examples, the system can include a pressure dampener, wherein the pressure dampener is configured to reduce pulsation in the fluid circuit caused by the pump. In some examples, the pump is a peristaltic pump. In some examples, the transport container further includes: a plurality of posts; and an organ retainer adjustably couplable to the plurality of posts. In some examples, the transport container further includes a one-way valve in fluid communication with the fluid circuit, the one-way valve configured to take in preservation fluid from the transport container when fluid is drawn out of the fluid circuit. In some examples, the system can include an accumulation chamber configured to seal to a port on a lid of the transport container, the accumulation chamber including a balloon configured to expand when the preservation fluid expands and contract when the preservation fluid contracts. In some examples, the system can include a bubble trap integrated in the fluid circuit.

In some examples, the systems for hypothermic transport of a biological sample described herein can include: a transport container configured to receive an organ, the transport container configured to be at least partially filled with preservation fluid; a pump configured to pump the preservation fluid from the transport container to the organ in a fluid circuit, the pump configured to operate independently of vessel resistance; and a flow regulator valve in fluid communication with the fluid circuit, the flow regulator valve configured to regulate a flow rate from the fluid circuit to the organ based on at least one organ parameter.

In some examples, the at least one organ parameter includes at least one of vessel resistance, organ temperature, or flow rate through the organ. In some examples, the transport container further includes an organ rest configured to support the organ. In some examples, the system can include an organ adapter configured to seal a vessel of the organ, the organ adapter adjustable along a groove of the organ rest. In some examples, the system can include a pressure dampener, wherein the pressure dampener is configured to reduce pulsation caused by the pump. In some examples, the pump is a peristaltic pump. In some examples, the transport container further includes: a plurality of posts; and an organ retainer adjustably couplable to the plurality of posts. In some examples, the transport container further includes a one-way valve in fluid communication with the fluid circuit, the one-way valve configured to take in preservation fluid from the transport container when fluid is drawn out of the fluid circuit. In some examples, the system can include a lid including a fill port and a vent port. In some examples, the system can include an accumulation chamber configured to seal to at least one of the fill port and the vent port, the accumulation chamber including a balloon configured to expand when the preservation fluid expands and contract when the preservation fluid contracts.

In some examples, the methods for hypothermic transport of a biological sample described herein can include: sealing a biological sample adapter to a vessel of a biological sample; placing the biological sample in a transport container; sealing a lid on the transport container; filling the transport container at least partially with preservation fluid through a fill port in the lid; and activating a pump, the pump configured to pump the preservation fluid from the transport container to a flow regulator valve and the biological sample adapter, the flow regulator valve configured to regulate a flow rate into the biological sample based on at least one biological sample parameter, and the pump configured to operate independently of the at least one biological sample parameter.

In some examples, the at least one biological sample parameter includes at least one of vessel resistance, biological sample temperature, or flow rate through the biological sample. In some examples, the method can include placing the biological sample on a biological sample rest. In some examples, the method can include coupling the biological sample adapter to a cannula receiver on the biological sample rest. In some examples, the method can include adjusting the biological sample adapter along a groove of the biological sample rest. In some examples, the method can include locking the biological sample adapter in place along the groove such that the vessel of the biological sample is under tension. In some examples, the method can include attaching a biological sample retainer to a plurality posts in the transport container such that the biological sample retainer secures the biological sample. In some examples, the method can include venting the transport container of fluid through a vent port. In some examples, the method can include attaching an accumulation chamber on at least one of the fill port or the vent port, the accumulation chamber including a balloon configured to expand when the preservation fluid expands and contract when the preservation fluid contracts. In some examples, the method can include preserving the biological sample for greater than 5 hours without adjusting parameters of the pump.

In some examples, the methods for hypothermic transport of a biological sample described herein can include: pumping, using a pump, preservation fluid from a transport container to an inlet of a fluid circuit at a rate independent of vessel resistance of a biological sample; pumping, using the pump, the preservation fluid from the inlet to a flow regulator valve; regulating, using the flow regulator valve, a flow rate into the biological sample based on at least one biological sample parameter; and pumping, using the pump, the preservation fluid from the flow regulator valve to a biological sample adapter in fluid communication with the biological sample.

In some examples, the at least one biological sample parameter includes at least one of vessel resistance, biological sample temperature, or flow rate through the biological sample. In some examples, the method can include applying tension to a vessel of the biological sample with the biological sample adapter while the biological sample is on a biological sample rest. In some examples, the method can include expanding a balloon in an accumulation chamber when the preservation fluid expands, wherein the accumulation chamber is attached to a port of a lid of the transport container. In some examples, the method can include cooling the biological sample with phase change material on an outer lid sealed to the transport container. In some examples, the method can include reducing pulsation in the fluid circuit with a pressure dampener. In some examples, the method can include measuring a pressure in the fluid circuit. In some examples, the method can include displaying the pressure in the fluid circuit on a display of the transport container. In some examples, the method can include measuring a temperature in the fluid circuit. In some examples, the method can include displaying the temperature in the fluid circuit on a display of the transport container.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1**A is a side, cross-sectional view of an example of an organ preservation apparatus.

FIG. **1**B is a top, angled view of the example of the organ preservation apparatus of FIG. **1**A with the inner lid uncoupled from the canister.

FIG. **1**C is a top, angled view of the example of the organ preservation apparatus of FIG. **1**A with the inner lid and outer lid coupled to the canister.

FIG. **2**A is a diagram of an example perfusion circuit for perfusing a biological sample.

FIG. **2**B is a schematic of the example perfusion circuit for perfusing a biological sample of FIG. **2**A.

FIG. **3**A shows an exploded view of an example of the outer lid, inner lid, inner canister, and inner canister.

FIG. **3**B shows a bottom perspective view of the example of the inner lid of FIG. **3**A.

FIG. **3**C illustrates the example of the outer canister of FIG. **3**A.

FIG. **3**D illustrates the example of the outer canister and outer lid of FIG. **3**A.

FIG. **4**A illustrates an example of a transporter for the apparatus of FIG. **1**C.

FIG. **4**B illustrates the example of the transporter body of FIG. **4**B with the transporter body partially transparent.

FIGS. **5**A-**5**D illustrate an example of an accumulation chamber.

FIGS. **6**A and **6**B illustrate the example of the accumulation chamber of FIG. **5**A on the vent port of the apparatus of FIG. **1**C.

FIG. **7**A illustrates an example of a kidney rest with posts and an organ retainer.

FIG. **7**B-**7**C illustrate an example of the kidney rest of FIG. **7**A.

FIG. **8**A illustrates an example of a round-hole cannula in an open configuration.

FIG. **8**B illustrates the example of the round-hole cannula in a closed configuration.

FIG. **9**A illustrates an example of an oval-hole cannula in an open configuration.

FIG. **9**B illustrates the example of the oval-hole cannula in a closed configuration.

FIG. **10** shows an example of an organ retainer.

FIG. **11** is a graph of kidney pressure and vascular resistance in an example of the organ preservation device described herein.

FIG. **12** is a graph of kidney pressure and perfusion flow in an example of the organ preservation device described herein.

FIG. **13**A shows an example of a flow regulator valve in the closed configuration.

FIG. **13**B shows the example of the flow regulator valve of FIG. **13**A in the partially open configuration.

FIG. **13**C shows the example of the flow regulator valve of FIG. **13**A in the open configuration.

FIG. **14** shows a cross-sectional view of an example of an organ preservation apparatus with a canister bubble trap and a perfusion line bubble trap.

FIG. **15** shows an example of an organ preservation apparatus for preserving an organ.

FIG. **16** shows an example of transporters aligned for stacking.

FIG. **17** shows an example of a transporter aligned for placement in a package.

FIG. **18**A shows an example of a removable electronics panel.

FIG. **18**B shows the example of the electronics panel of FIG. **18**A being positioned on the transporter.

DETAILED DESCRIPTION

The disclosed systems for hypothermic transport of samples provide a sterile, temperature-stabilized environment for transporting samples while providing an ability to perfuse organs with a constant pump system. Because of these improvements, users of examples of the systems described herein can reliably transport samples over much greater distances, thereby substantially increasing the pool of available tissue donations. Additionally, because the tissues may be in better condition upon delivery, the long-term prognosis for the recipient may be improved.

In examples, the systems described herein may use a pump to perfuse one or more organs with fluid from within the canister in a cycle. The pump can operate at a constant rate so that preservation fluid is consistently being circulated throughout the system in a closed loop. A fluid channel may carry the fluid from the chamber to the pump and then to the organ. The organ can release the fluid back into the chamber. A flow regulator valve or manifold in line with the fluid channel can allow some fluid to be released into the chamber before it reaches the organ when the resistance of the cannulated organ is above a threshold. For instance, the resistance of the renal artery of the kidney can determine how much fluid is released from the manifold as opposed to how much enters the kidney.

The pump can be a peristaltic pump in contact with the fluid channel or any other suitable pump. The point of contact with the pump can be outside the canister containing the organ. Once the pump is turned on, the system can operate without any inputs or outputs during transport. The same fluid can be circulated throughout transportation or storage. The system can include a pressure dampener that reduces pulsation of the flow caused by the pump.

Hypothermic transport systems such as those described herein can comprise a self-purging preservation apparatus and an insulated transport container. The self-purging preservation apparatus may receive the tissue for transport, and keep it suspended or otherwise supported in a surrounding pool of preservation solution. In some examples, the preservation solution can be chilled to around 4° C. The preservation solution may be chilled to below 4° C. The self-purging preservation apparatus may comprise a number of configurations suitable to transport tissues hypothermically.

In some embodiments, a transport device may be configured to self-purge excess fluid (e.g., liquid and/or gas). For example, in some embodiments, such a device includes a lid assembly in which at least a portion of the lid assembly is inclined with respect to a horizontal axis. The inclined portion of the lid assembly may be configured to facilitate the flow of fluid towards a vent port disposed at substantially the highest portion of a chamber of the lid assembly. In this manner, excess fluid can escape the device via the purge port. Also in this manner, when excess liquid is expelled from the device via the purge port, an operator of the device can determine that any excess gas has also been purged from the device, or at least from within a tissue chamber of the device, because the gas is lighter than the liquid and will move towards and be expelled via the purge port before excess liquid.

As used in this specification, the singular forms "a," "an" and "the" include plural referents unless the context clearly dictates otherwise. Thus, for example, the term "a fluid" is intended to mean a single fluid or a combination of fluids.

As used herein, "a fluid" refers to a gas, a liquid, or a combination thereof, unless the context clearly dictates otherwise. For example, a fluid can include oxygen, carbon dioxide, or another gas. In another example, a fluid can include a liquid. Specifically, the fluid can be a liquid perfusate. In still another example, the fluid can include a liquid perfusate with a gas, such as oxygen, mixed therein or otherwise diffused therethrough.

As used herein, "tissue" refers to any tissue of a body of a patient, including tissue that is suitable for being replanted or suspected of being suitable for replantation. Tissue can include, for example, muscle tissue, such as, for example, skeletal muscle, smooth muscle, or cardiac muscle. Specifically, tissue can include a group of tissues forming an organ, such as, for example, the skin, lungs, cochlea, heart, bladder, liver, kidney, or other organ. In another example, tissue can include nervous tissue, such as a nerve, the spinal cord, or another component of the peripheral or central nervous system. In still another example, tissue can include a group of tissues forming a bodily appendage, such as an arm, a leg, a hand, a finger, a thumb, a foot, a toe, an ear, genitalia, or another bodily appendage. While the systems are described as relating to the transport of tissues, such as organs, it is also envisioned that the systems could be used for the transport of body fluids, which may be held in another container within the self-purging preservation apparatus. Body fluids may include blood and blood products (whole blood, platelets, red blood cells, etc.) as well as other body fluids for preservation.

FIG. **1**A is a cross-sectional, side view of an example of an organ preservation apparatus. FIG. **1**B is a top, angled view of the example of the organ preservation apparatus of FIG. **1**A with the inner lid **40** uncoupled from the canister **38**. FIG. **1**C is a top, angled view of the example of the organ preservation apparatus of FIG. **1**A with the inner lid **40** and outer lid **42** coupled to the canister **38**.

The apparatus **10** can be configured to store a biological sample T. The apparatus **10** can be configured to perfuse the biological sample T using a constant pump system, as described with respect to FIGS. **2**A and **2**B. Pumping preservation fluid in the biological sample T can extend the preservation of the sample. The biological sample T can be an organ, for example a kidney. In other embodiments, the organ can be a lung, a heart, or a liver. The constant pump system can be advantageous to allow flow through the biological sample T without requiring changes to the pump rate. The constant pump system, once activated, can operate without intervention throughout transportation.

The apparatus **10** can include an inner lid **40** and a canister **38** that can be coupled using clamps **56**. An outer lid **140** and outer canister **138** can contain the canister **38** and inner lid **40** as described with respect to FIG. **3**A. The outer lid **140** and outer canister **138** can be coupled using clamps **156**.

The biological sample T can be positioned on an organ rest **18**, for example an organ rest as described with respect to FIGS. **7**B and **7**C. The biological sample T can be connected to a cannula **26**, as described with respect to FIGS. **8**A, **8**B, **9**A, and **9**B. The cannula **26** can be connected to a cannula receiver **20**. The cannula receiver **20** can be positioned in a groove **21** of the organ rest. The pumping chamber **16** beneath the organ rest **18** can be filled with fluid, for example preservation fluid. The preservation fluid can be pumped from the pumping chamber **16** to an inlet **12**. The entire canister **38** can be substantially filled with preservation fluid. The inlet **12** can include a filter as described with respect to FIG. **2**B.

In certain examples, the inner lid **40** can include a fill port **28** and a vent port **30**. The canister **38** can be filled with preservation through into the fill port **28**, for example until the canister **38** is substantially filled. Gas and fluid can exit the canister from the vent port **30** while the canister **38** is being filled. The fill port **28** can be configured to permit fluid (e.g., perfusate) to be introduced to the canister **38**. In this manner, fluid can be introduced into the canister **38** as desired by an operator of the apparatus. For example, in some embodiments, a desired amount of perfusate is introduced into the canister **38** via the fill port **28**, such as before disposing the tissue T in the canister **38** and/or while the biological sample T is received in the tissue chamber. In some embodiments, the fill port **28** is a unidirectional port, and thus is configured to prevent the flow of fluid from the canister **38** to an area external to the tissue chamber through the port. In some embodiments, the fill port **28** includes a luer lock. The canister **38** may be of any suitable volume necessary for receiving the tissue T and a requisite amount of fluid for maintaining viability of the tissue T. In one embodiment, for example, the volume of the canister **38** is approximately 2 liters. Once the canister **38** is substantially filled with preservation fluid, the fill port **28** and vent port **30** can be sealed. After filling, an accumulation chamber can be positioned on the fill port **28**, as described with respect to FIGS. **6**A and **6**B.

Fluid, for example gas, in the canister **38** can be vented through the vent port **30** and/or the fill port **28**. The vent port **30** can be configured to permit fluid to flow into the atmosphere external to the apparatus **10**. In some embodiments, the vent port **30** is configured for unidirectional flow, and thus is configured to prevent a fluid from being introduced into the canister **38** via the port (e.g., from a source external to the apparatus **10**). In some embodiments, the vent port **30** includes a luer lock. After venting, an accumulation chamber can be positioned on the vent port **30**, as described with respect to FIGS. **6**A and **6**B. The accumulation chamber can minimize impacts of external forces or pressures impacting the pressure of the fluid.

In some examples, when the inner lid **40** is coupled with the canister **38**, the inlet **12** can be in fluid communication with the lid inlet **44**. The lid inlet **44** can be in fluid communication with an outer tube **36**. The outer tube **36** can be in contact with a pump. The outer tube **36** can be in fluid communication with a lid outlet **46**. When the inner lid **40** is coupled with the canister **38**, the lid outlet **46** can be in communication with the outlet **14**. The outlet **14** can include a flow regulator valve **48**, or a pressure relief valve. The outlet can be in fluid communication with the cannula receiver **20**. A pressure sensor **52** and pressure damper **54** can be positioned near the outer tube **36** in the inner lid **40**. A temperature sensor **50** can be positioned near on the bottom of the inner lid **40** near the cannula receiver **20** when the inner lid **40** and the canister **38** are coupled. The temperature sensor **50** can measure the temperature of the preservation fluid in the canister **38**. Fluid can flow from the outlet **14** to the biological sample T, for example through the cannula receiver **20** and the cannula **26**. Fluid can flow into the biological sample T through a vessel such as a renal artery of a kidney. Fluid can flow out of the biological sample through a vessel such as a renal vein or ureter of a kidney. After exiting the biological sample T, the fluid can collect in the organ chamber **17**.

As shown in FIG. **1**A, the organ rest **18** can be inclined at an angle. The organ rest **18** can incline from a lower point on one side **22** to a higher point on the other side **24**. The incline of the organ rest **18** can be advantageous to maintain tension on a cannulated vessel of the biological sample T. The cannulated vessel can be an artery, for example a renal artery. In some embodiments, the cannulated vessel can be a renal vein or ureter. The organ rest **18** can support the biological sample. The organ rest **18** can be curved to accommodate a bottom surface of the biological sample T.

In some examples, the organ chamber **17** can be formed by the canister **38** and the organ rest **18** above the pumping chamber **16**. The organ rest **18** can be non-permeable and can have apertures **58** to allow fluid to move between the organ chamber **17** and the pumping chamber **16**.

In some examples, the organ rest **18** can be integral with the bottom of the canister **38** or removably connected to the canister **38**. The organ rest **18** can have a groove **21**. The groove **21** can be a narrow opening with a raised portion surrounding it. The cannula receiver **20** can be slidably positioned in the groove **21**, such that the device is adjustable. The cannula receiver **20** can be locked into a position along the groove **21** based on the length of the cannulated vessel, for example the length of the renal artery. The cannula receiver **20** can be positioned such that there is tension on the cannulated vessel when the biological sample T is secured on the organ rest **18**. A user can connect the cannula **26** to the biological sample T, connect the cannula **26** to the cannula receiver **20**, move the cannula receiver **20** along the groove **21** to adjust for vessel length, and lock the cannula receiver **20** in place. The cannula receiver **20** and/or the cannula **26** can be referred to as an organ adapter, for example a kidney adapter.

In some examples, the organ rest **18** can have a retention mechanism to secure the biological sample T in place. This can prevent lateral and vertical movement of the biological sample T during transportation. The retention mechanism can be any suitable device such as, for example, a net, a cage, a sling, a strap, or the like. The retention mechanism can be attached to posts on the organ rest, as described with respect to FIG. **7**A. In some embodiments, the apparatus **10** includes a basket (not shown) or other support mechanism configured to support the tissue T when the tissue T is on the organ rest **18** or otherwise received in the apparatus **10**.

In certain examples, the sterile canister can be constructed from a sterilizable material, i.e., made of a material that can be sterilized by steam (autoclave), chemically, or via UV irradiation, or another suitable form of sterilization. Sterilization can prevent tissues from becoming infected with viruses, bacteria, etc., during transport. In a typical embodiment, the sterile canister can be delivered in a sterile condition and sealed in sterile packaging. In some embodiments, the sterile canister apparatus will be re-sterilized prior to reuse, for example at a hospital. In other embodiments, the sterile canister will be disposable.

In examples, the systems, methods, and apparatuses described herein for hypothermic transport of tissues described herein may provide for the transport of biological samples (e.g., tissue, organs, or body fluids) over long distances and time periods while maintaining a temperature of 4-8° C., or in another example, 2-10° C. For example, temperature may be controlled for at least about 6 h, 12 h, 24 h, 48 h, or more time. Systems of the disclosure can enable medical professionals to keep tissues (e.g., organs) in a favorable hypothermic environment for extended periods of time, thereby allowing more time between harvest and transplant.

In some examples, in use, the biological sample T is coupled to the cannula **26** and secured on the organ rest **18**. A desired amount of perfusate is introduced into the canister **38** via the fill port **28**. The canister **38** can be filled with the perfusate such that the perfusate volume rises to the highest portion of the tissue chamber. The canister **38** can be filled with an additional amount of perfusate such that the perfusate flows from the organ chamber **17** through the apertures **58** into the pumping chamber **16**. The canister **38** can continue to be filled with additional perfusate until all atmospheric gas that initially filled the canister **38** rises and escapes through the vent port **30**. Because the gas will be expelled via the vent port **30** before any excess perfusate is expelled (due to gas being lighter, and thus more easily expelled, than liquid), an operator of the apparatus **10** can determine that substantially all excess gas has been expelled from the pumping chamber when excess perfusate is released via the port. Due to the release of the excess gas, in certain examples, the apparatus **10** can be characterized as self-purging. When perfusate begins to flow out of the vent port **30**, the apparatus **10** is in a "purged" state (i.e., all atmospheric gas initially within the canister **38** has been replaced by perfusate). When the purged state is reached, the operator can close both the fill port **28** and vent port **30**, preparing the apparatus **10** for operation.

The canister **38** can be constructed of any suitable material. In some embodiments, the canister **38** is constructed of a material that permits an operator of the apparatus **10** to view at least one of the tissue T or the perfusate received in the organ chamber **17**. For example, in some embodiments, the canister **38** is substantially transparent. In another example, in some embodiments, the canister **38** is substantially translucent. Lastly, in certain examples, the canister may be opaque to obscure viewing of the tissue. The organ chamber **17** can be of any suitable shape and/or size. For example, in some embodiments, the organ chamber **17** can have a perimeter that is substantially oblong, oval, round, square, rectangular, cylindrical, or another suitable shape. The canister **38** and organ rest **18** can be plastic. The apparatus **10** can be easily transportable.

In some examples, the flow regulator valve **48** can release fluid from the fluid channel rather than allowing it to flow to the biological sample T when the resistance of the cannulated vessel is above a threshold. Once fluid enters the flow regulator valve **48**, it can flow either directly to the biological sample or back into the organ chamber **17**. The flow regulator valve **48** can be a mechanical device that releases fluid when a certain pressure is exceeded. The flow regulator valve **48** can close more when pressure decreases such that increased flow is directed to the biological sample T. The flow regulator valve **48** can be variable such that when pressure or renal resistance decreases, more flow is directed to the kidney. Resistance in the cannulated vessel can be higher, for example, when the organ is exposed to low temperatures before transport. A higher resistance of the vessel of the kidney can result in higher pressure. Setting the relief valve **48** to allow an optimal or enhanced flow of fluid into the cannulated vessel can obviate the need to change pump parameters during transportation. The flow regulator valve **48** can be metal, for example stainless steel. In some embodiments, the flow regulator valve **48** may be pre-set at a pressure threshold or series of pressure thresholds. In another embodiment, the flow regulator valve **48** may be adjustable, allowing for a user to set pressure thresholds.

In some examples, when pressure is below the threshold, the fluid can be allowed to flow into the biological sample T. For high renal resistance, the pressure threshold can be 65 mmHg. In some embodiments, for high renal resistance, the pressure threshold can be 60-70 mmHg. In some embodiments, for high renal resistance, the pressure threshold can be 50-80 mmHg. For medium renal resistance, the pressure threshold can be approximately 30-65 mmHg. In some embodiments, for medium renal resistance, the pressure threshold can be approximately 10-100 mmHg. For low renal resistance, the pressure threshold can be approximately 30 mmHg. In some embodiments, for low renal resistance, the pressure threshold can be approximately 10-50 mmHg. In some examples, the device can maintain a pressure operating range of 0 mmHg to 50 mmHg with a flow operating range of 0 mL/min to 150 mL/min.

In certain examples, high renal vascular resistance can mean a perfusion pressure of greater than approximately 35 mmHg. In some examples, medium renal vascular resistance can mean a perfusion pressure of between approximately 25 and approximately 35 mmHg. In some examples, low renal vascular resistance can mean a perfusion pressure of less than approximately 25 mmHg.

The flow regulator valve **48** can operate such that flow rate to the kidney varies based on pressure and/or renal resistance. Renal resistance can be measured in units of mmHg*min/mL. In some embodiments, when pressure is greater than 42 mmHg and/or renal resistance is greater than 1.55, the flow rate to the kidney can be less than or equal to 30 mL/min. In some embodiments, when pressure is greater than 30-50 mmHg and/or renal resistance is greater than 0.5-2.5, the flow rate to the kidney can be less than or equal to 10-50 mL/min. In some embodiments, when pressure is approximately 42 mmHg and/or renal resistance is approximately 1.39, the flow rate to the kidney can be approximately 30 mL/min. In some embodiments, when pressure is approximately 30-50 mmHg and/or renal resistance is approximately 0.5-2.5, the flow rate to the kidney can be approximately 10-50 mL/min. In some embodiments, when pressure is between than 23 mmHg and 42 mmHg and/or renal resistance is between 0.16 and 1.55, the flow rate to the kidney can be between 30 and 130 mL/min. In some embodiments, when pressure is between than 5 mmHg and 75 mmHg and/or renal resistance is between 0.01 and 5, the flow rate to the kidney can be between 10 and 200 mL/min. In some embodiments, when pressure is less than 23 mmHg and/or renal resistance is less than 0.16, the flow rate to the kidney can be greater than or equal to 130 mL/min. In some embodiments, when pressure is less than 5-50 mmHg and/or renal resistance is less than 0.01-5, the flow rate to the kidney can be greater than or equal to 75-200 mL/min. In some embodiments, when pressure is approximately 23 mmHg and/or renal resistance is approximately 0.18, the flow rate to the kidney can be approximately 130 mL/min. In some embodiments, when pressure is approximately 5-50 mmHg and/or renal resistance is approximately 0.01-5, the flow rate to the kidney can be approximately 75-200 mL/min.

Renal resistance, or resistance of the cannulated vessel, can be determined by dividing the pressure at the flow regulator valve **48** by the flow rate into the biological sample. The flow regulator valve **48** can control flow to the biological sample, for example the renal artery of a kidney, based on the non-linear equation below.

$$P = A - BF - CF^2$$

In the equation above, P represents pressure at the flow regulator valve **48** and F represents flow rate into the biological sample. A, B, and C are constants. In one example, A can be 42.93, B can be 0.01076, and C can be 0.001072. In some examples, A can be between 35 and 45, B can be between 0.005 and 0.05, and C can be between 0.0005 and 0.0015. In some embodiments, A can be between 25 and 75, B can be between 0.00005 and 0.1, and C can be between 0.00005 and 0.1. In some embodiments, A can be between 10 and 100, B can be between 0.000005 and 1, and C can be between 0.000005 and 1. In some embodiments, the equation used to relate pressure at the flow regulator valve **48** to the flow rate into the organ can be binomial or polynomial.

In some examples, the flow regulator valve **48** can be set to control flow rate into the organ based on the equation above by choosing a valve with a particular cracking pressure threshold. One of skill in the art will understand that a cracking pressure threshold can be the minimum pressure required to open the valve and allow fluid or gas flow. The cracking pressure threshold can be the point at which the valve begins to open and initial resistance is overcome, such that a minimum amount of fluid is released from the fluid channel. The cracking pressure threshold can correspond to the pump speed such that the flow rate into the organ corresponds to the equation described herein. The flow regulator valve **48** in this embodiment can be based on a pump with a speed of 130 mL/min. The flow regulator valve **48** in this embodiment can be based on a pump with a speed of 100-150 mL/min. The flow regulator valve **48** in this embodiment can be based on a pump with a speed of 50-200 mL/min. The flow regulator valve **48** in this embodiment can be based on a pump with a speed of 10-500 mL/min. The flow regulator valve **48** can be a spring-loaded ball check valve, for example with a single ball, in which the spring allows preservation fluid to escape the fluid circuit into the organ chamber **17** when pressure is above the cracking pressure threshold. This pressure increase can be caused by arterial resistance, such as renal resistance, slowing flow into the organ and causing build up in the valve. Once cracking pressure is reached, higher pressures can cause the spring to constrict non-linearly, causing the flow from the flow regulator valve **48** to the organ chamber **17** to be non-linear in relation to the renal resistance. Because flow from the flow regulator valve **48** to the organ chamber **17** is non-linear in this case, the flow from the flow regulator valve **48** to the organ can also be non-linear in relation to the renal resistance, based on the amount of fluid exiting the fluid circuit. In certain examples, the use of the flow regulator valve **48** can allow the constant pump to be effective throughout transport by controlling the flow rate into the organ based on the renal resistance. The flow rate caused by the flow regulator valve **48** can be enhanced or even optimal, such that the flow does not damage the artery by overcoming resistance too harshly nor does the flow rate allow too little flow in when renal resistance would allow more flow without causing damage.

The flow rate to the kidney can be determined based on the pressure and/or the renal resistance in Table 1, below. The pressure measurements in Table 1 can be within approximately 5 mmHg. The renal resistance values in Table 1 can be within approximately 15%. The flow rate measurements can be within approximately 5 mL/min.

TABLE 1

Pressure (mmHg)	Flow Rate (mL/min)	Renal Resistance (mmHg*min/mL)
>42	≤30	>1.55
42	30	1.39
41	35	1.18
40	45	0.90
39	55	0.71
38	60	0.64
37	70	0.53
36	75	0.48
35	80	0.44
34	85	0.40
33	90	0.37
32	95	0.34
31	100	0.31
30	105	0.29
29	110	0.26
28	115	0.24
27	118	0.23
26	120	0.22
25	125	0.20
24	128	0.19
23	130	0.18
<23	≥130	<0.16

In some examples, the temperature sensor **50** can be configured to measure the temperature of the organ chamber **17**. The temperature sensor **50** can be positioned on the bottom of the inner lid **40**. When the canister **38** is filled with fluid, the fluid can contact the temperature sensor **50**. The temperature sensor **50** can measure the temperature of the preservation fluid. In alternate embodiments, the temperature sensor **50** can be positioned on the organ rest **18**, in the pumping chamber **16**, on the outer tube **36**, on the cannula receiver **20**, or on the bottom of the inner lid **40** above the biological sample T. The temperature of the biological sample T can be measured directly by the temperature sensor **50** or indirectly based on the temperature measurement of the fluid. The temperature sensor **50** can communicate the measurement to an external device or a display on the device.

In some examples, a pressure sensor **52** and pressure dampener **54** can be positioned near the tube **36**, for example within the inner lid **40**. The pressure sensor **52** can measure the pressure of the fluid in the fluid channel. The pressure sensor **52** can be in-line with the perfusion circuit. The pressure sensor **52** can be connected to the pressure dampener **54**. The pressure dampener **54** can have three openings. Two openings of the pressure dampener **54** can be connected to the fluid circuit and one opening of the pressure dampener can be connected to the pressure sensor **52**. The pressure dampener **52** can include a mechanism to reduce pulsation caused by the pump. The pressure sensor **52** can communicate the measurement to an external device or a display on the device. The pressure dampener **54** can be a pulsation dampener that reduces the pulsation caused by the pump. The apparatus **10** can include a cooling element, for example eutectic cooling packs, in proximity to the biological sample T.

In certain examples, the apparatus **10** can be a canister **38** and lid **40** contained inside an outer canister **138** and outer lid **140**. The apparatus **10** can be contained inside a shipper as described with respect to FIGS. **4**A and **4**B. The apparatus **10** can be made of plastic, for example polycarbonate.

In some examples, the apparatus **10** can be used to perfuse a liver. In some examples, the organ can be perfused via the portal vein and hepatic artery. In some examples, the organ can be perfused via the portal vein only. In some examples, the organ can be perfused using a barbed or straight cannula.

FIG. **2**A is a diagram of an example perfusion circuit **200** for perfusing a biological sample T. FIG. **2**B is a schematic of the example perfusion circuit **200** for perfusing a biological sample T of FIG. **2**A.

In some examples, the pump **202** can facilitate flow of the preservation fluid. The pump **202** can be a peristaltic pump in contact with the outer tube **36**. The pump **202** can be positioned on the shipper, as described with respect to FIGS. **4**A and **4**B. As the peristaltic pump **202** rolls in contact with the flexible outer tube **36**, the force applied to the outer tube **36** can cause fluid to flow through the perfusion circuit **200**. This can cause the inlet **12** to draw fluid from the pumping chamber **16**. The pump **202** can be a small footprint flip-top pump. A user can thread the tube **36** into the pump **202**.

In certain examples, a fluid filter **204** can filtrate the fluid. The fluid filter **204** can be an in-line filter at the pump inlet **12** or in the fluid channel between the inlet **12** and the cannula receiver **20**. In some embodiments, the filter **204** can be at another point in the fluid channel. The fluid filter can be a 20-micron stainless steel filter in a polyethylene housing.

As shown with respect to FIG. **2**B, the inlet **12** can take in fluid from the organ container **38**. Fluid can flow from the inlet **12** to the outer tube **36** in contact with the pump **202**, for example by traveling through the lid inlet **44** and the outer canister inlet **144**. Fluid can then flow from the outer tube **36** to the outlet **14**, for example through the lid outlet **46** and the outer canister outlet **146**. Fluid can flow from the outlet **14** to cannula **26** and/or the flow regulator valve **48**, depending on the renal resistance. Fluid can flow from the cannula **26** to the biological sample T, and then from the biological sample T into the storage chamber **17**. As the peristaltic pump **202** rolls, it can enact force on the outer tube **36** in a pulsatile manner. The pressure dampener **54** can cause flow through the perfusion circuit **200**, or fluid circuit, to become more smooth or constant as opposed to pulsatile. In some examples, the perfusion circuit **200** can perfuse the organ in a pulsatile manner. For example, the pulsatile perfusion may involve repeated perfusion cycles with a period of time in between.

In certain examples, an operator can remove fluid, for example gas, from the perfusion circuit **200** at the purge point **206**. The operator can use a syringe, for example, to draw fluid from the purge point **206**. The one-way valve **208** can allow preservation fluid from the canister **38** to backfill the perfusion circuit **200** when fluid is drawn out of the purge point **206**. Advantageously, this can allow the perfusion circuit **200** to be purged of gas. The purge point **206** can prevent fluid from entering or exiting the perfusion circuit **200** when fluid is not being drawn out. The one-way valve **208** can prevent fluid from entering or exiting the perfusion circuit **200** when fluid is not being drawn out.

In some examples, the pump **202** can operate at a constant rate throughout transport. The pump **202** can operate independently of renal resistance. For example, the pump **202** can vary in speed based on hardware or software, but the pump may not change its speed based on the renal resistance or pressure in the perfusion circuit **200**. The perfusion circuit **200** can be a closed loop such that the volume of preservation fluid is constant. Advantageously, the perfusion circuit **200** can operate continuously without intervention. A battery can power the pump **202** with a constant voltage. A battery can power the pump **202** with a voltage independent of renal resistance. Renal resistance can mean the arterial resistance or the resistance of a vessel of the biological sample.

FIG. **3**A shows an exploded view of an example of the outer lid **140**, inner lid **40**, inner canister **38**, and outer canister **138**. FIG. **3**B shows a bottom perspective view of the example of the inner lid of FIG. **3**A. FIG. **3**C illustrates the example of the outer canister **138** of FIG. **3**A. FIG. **3**D illustrates the example of the outer canister **138** and outer lid **140** of FIG. **3**A.

In some examples, the inner canister **38** can couple to the inner lid **40** such that the lid inlet **44** and lid outlet **46** couple with the inlet **14** and outlet **16**, respectively. The clamps **56** can mechanically secure the inner canister **38** to the inner lid **40**.

In some examples, the inner lid **40** can couple to the outer canister **138**, for example after the lid **40** and the inner canister **38** have been coupled. The lid inlet **44** can be fluidically connected to the outer inlet connector **344**. The lid outlet **46** can be connected to the outer outlet connector **346**. The inner lid **40** can connect to the outer canister **138** such that the outer inlet connector **344** connects to the outer canister inlet **144** and the outer outlet connector **346** connects to the outer canister outlet **146**. The electrical connector **364** on outer canister **138** can connect to the electrical receiver **362** on the inner lid **40**. The electrical receiver **362** can be positioned between the outer inlet connector **344** and the outer outlet connector **346**. The electrical connector **364** can be positioned between the outer canister inlet **144** and the outer canister outlet **146**.

As shown in FIG. **3**D, the outer canister **138** can couple with the outer lid **140**. The clamps **156** can mechanically secure the outer canister **138** to the outer lid **140**. The purge point **206** can have a cap to prevent fluid from exiting the circuit. The outer canister **138** and outer lid **140** can protect the apparatus **10** from the environment, ensuring that the biological sample remains sterile.

FIG. **4**A illustrates an example of a transporter **400** for the apparatus **10**. FIG. **4**B illustrates the example of the transporter body **438** for the apparatus **10** of FIG. **4**B with the transporter body **438** partially transparent.

The transporter **400** can have a transporter body **438** and a transporter lid **440**. In some examples, the transporter lid **440** can be coupled to the transporter body and mechanically secured using clamps **456**. The pump **202** can be positioned on the transporter body **438**. The pump **202** can operate at 7.2 V. In some embodiments, the pump **202** can operate at 1-10 V. In some embodiments, the pump **202** can operate at 1-20 V. The battery in the transporter **400** can supply the voltage at which the pump **202** operates. The transporter can include a button or switch for turning the pump **202** on and off. The pump **202** can generate a flow rate of approximately 130 mL/min. In some embodiments, the pump **202** can generate a flow rate of 110-150 mL/min. In some embodiments, the pump **202** can generate a flow rate of 80-180 mL/min. In some embodiments, the pump **202** can generate a flow rate of 50-200 mL/min. Advantageously, the use of a battery at approximately 7.2 V, or within the disclosed ranges, to power the pump **202** with a flow rate of approximately 130 mL/min, or within the disclosed ranges, can efficiently perfuse the organ.

In certain examples, the transporter **400** can have a handle and wheels to facilitate transport. The handle can be a telescoping handle. The transporter **400** can include a left and right lifting handle for lifting the shipper during transportation. The transporter **400** can include a lock-ring for securing the apparatus **10** in place during transportation. The transporter **400** can include an electrical connector **474** for connection to the apparatus. The electrical connector **474** can connect the apparatus **10** to the display **470**.

In certain examples, the display **470** can include a datalogger. The display **470** can display temperature within the canister **38**, ambient temperature, pressure within the fluid circuit, battery voltage, and/or other parameters. The display **470** can be connected to the apparatus **10** through wired or wireless connection. The display **470** can be connected to the temperature sensor **50** and/or the pressure sensor **52**. The transporter **400** can include a battery or power source connected to the display **470** and/or the pump **202**.

The transporter **400** can contain cooling media to maintain the temperature of the biological sample. For example, the cooling media can be placed on top of the outer lid **140** and around the outer canister **138**. The cooling media can include eutectic cooling blocks. The cooling media can have a temperature of 1° C. In some embodiments, the cooling media can have a temperature of 0.5-10° C. The organ can be transported at a temperature with an operating range of 4-8° C. In some embodiments, rgan can be transported at a temperature with an operating range of 2-10° C.

Cooling media can be positioned around the outer canister **138** and/or on top of the outer lid **140**. The cooling media around the outer canister **138** can be ribbons of phase change material. The cooling media around the outer canister **138** can include five ribbons of phase change material. Three ribbons can be arranged radially around the outer canister **138** at or near the bottom of the outer canister **138**. Two ribbons can be arranged radially around the top of the outer canister **138** and/or radially around the outer lid **140**. In some examples, the cooling media around the outer canister **138** can include 1-10 ribbons of phase change material. In some examples, the cooling media around the outer canister **138** can include 1-20 ribbons of phase change material. The ribbon of phase change material can include individual phase change material packs that are linked together. The ribbons may be overlayed radially around other ribbons. In some examples, 1-5 ribbons can be arranged radially around the outer canister **138** at or near the bottom of the outer canister **138**. In some examples, 1-10 ribbons can be arranged radially around the outer canister **138** at or near the bottom of the outer canister **138**. In some examples, 1-5 ribbons can be arranged radially around the top of the outer canister **138** and/or radially around the outer lid **140**. In some examples, 1-10 ribbons can be arranged radially around the top of the outer canister **138** and/or radially around the outer lid **140**.

The cooling media on top of the outer lid **140** can be a pouch of phase change material. The pouch of phase change material can contain four individual phase change material envelopes. In some examples, the pouch of phase change material can contain 2-6 individual phase change material envelopes. In some examples, the pouch of phase change material can contain 1-10 individual phase change material envelopes. The pouch of phase change material can be positioned under the handle of the outer lid **140**. In some examples, the phase change material can have a temperature of 1° C., 2° C., 3° C., 4° C., 5° C., 6° C., 7° C., or 8° C.

In some examples, the transporter **400**, or shipper, can be made of plastic or a polymer, for example rigid, molded expanded polystyrene.

FIGS. **5**A-**5**D illustrate an example of an accumulation chamber **500**. The accumulation chamber **500** can compensate for fluid expansion within the canister **38** due to external pressure changes. The accumulation chamber **500** can include a top portion **582**, a bottom portion **584**, and a balloon **586** between the top portion **582** and the bottom portion **584**.

The balloon **586** can expand when fluid in the canister **38** expands due to high external pressure and contract when fluid in the canister contracts due to low external pressure. This is advantageous as it can allow the canister **38** to respond to changes in external pressure. The canister **38** can be filled with preservation fluid, so forces acting on the canister, transporter, or apparatus can cause the preservation fluid to expand. Without an accumulation chamber **500**, this could affect the perfusion of the organ. If the apparatus is jostled during transport or transported at an altitude that results in a pressure change, the accumulation chamber **500** can prevent the internal pressure from significantly changing, thus protecting the organ.

In some examples, the balloon **586** can be made of urethane. In some embodiments, the balloon **586** can be a flexible membrane or a series of flexible membranes. A space between the flexible membranes or between a flexible membrane and the top portion **582** or the bottom portion **584** of the accumulation chamber **500** can expand and contract in response to external pressure changes.

In some examples, the top portion **582** and bottom portion **584** can be made of plastic, for example the same plastic used in the apparatus **10** of FIG. **1**A-C. The top portion **582** can have apertures **588** to allow the balloon **586** to expand. The top portion **582** and bottom portion **584** can limit how much the balloon **586** expands. The top portion **582** can couple to the bottom portion **584** using clips **589**.

In some examples, the bottom portion **584** can have an accumulation chamber connector **587**. The accumulation chamber connector **587** can seal to the apparatus **10**, for example the fill port **28** and/or the vent port **30** of the inner lid **40**. The accumulation chamber connector **587** can be in fluid communication with the balloon **586**.

FIGS. **6**A and **6**B illustrate the example of the accumulation chamber **500** of FIG. **5**A on the vent port **30** of the apparatus **10**.

In certain examples, the accumulation chamber **500** can be positioned on the fill port **28** and/or the vent port **30**. The accumulation chamber **500** can be in fluid communication with the interior of the canister **38**. In some embodiments, the fill port **28** and vent port **30** can be sealed with caps and/or balloons. Advantageously, the accumulation chamber **500** allows some expansion of the fluid in the canister **38** to maintain pressure, but does not allow significant changes in volume of the canister **38** or density of the fluid.

FIG. **7**A illustrates an example of an organ rest **718** with posts **790** and an organ retainer **792**.

The organ rest **718**, or kidney rest, can be similar to the organ rest **18** of FIG. **1**A-C. In some examples, the posts **790** can be notched posts. The retainer **792** can have attachment mechanisms **794** that can fit along the notches. The retainer **792** can be moved up and down along the posts **790** to adjust for the size of the biological sample T. The retainer **792** can be locked in place once a desired height is determined, for example by sliding the retainer such that the attachment mechanisms **794** engage the notches. The retainer **792** can be couplable with the posts **790** at various points.

In some examples, the retainer **792** can have a plurality of straps configured to keep the biological sample T from moving horizontally and/or vertically. The straps can be made of rubber, plastic, or polymer. The straps can be curved to accommodate the shape of the biological sample T.

FIG. **7**B illustrates a side view of the example of the organ rest **718** of FIG. **7**A. FIG. **7**C illustrates a top view of the example of the organ rest **718** of FIG. **7**A.

In some examples, the organ rest **718** can include a groove **721**. A cannula receiver **720**, or organ adapter, can be slidable along the groove **721**. The cannula receiver **720** can lock in place along the groove **721**, for example by securing pins fixed to the cannula receiver **720** at a position along recesses around the groove **721**. A user can lock the cannula receiver **720** in a position such that an artery of the biological sample, for example a renal artery of a kidney, is in tension when the biological sample is cannulated or otherwise attached to an adapter. The cannula receiver **720** can be pulled out to a position in which the cannula receiver **720** can slide along the groove **721** and pushed into a position in which the cannula receiver **720** is locked in place.

FIG. **8**A illustrates an example of a round-hole cannula **800** in an open configuration. FIG. **8**B illustrates the example of the round-hole cannula **800** in a closed configuration.

The cannula **800** can be configured to seal to the artery, for example a renal artery. In some examples, the cannula **800** can include a base portion **802**. The base portion **802** can be divided into two portions, a first base portion **802*a*** and a second base portion **802*b***. The second base portion **802*b*** can be a slider portion that slides away from the first base portion **802*a*** to allow the vessel to be more easily pulled through the central aperture **804**. The second base portion **802*b***, or slider portion, can slide toward the first base portion **802*a*** to keep the vessel from moving significantly during transportation without sealing the vessel entirely in place. The first base portion **802*a*** can be moved toward and away from the second base portion **802*b***. The base portions **802*a,b*** can be moved away from each other to expand a central aperture **804** such that the vessel, artery, or artery cuff can be pulled through the central aperture **804**. Then, the base portions **802*a,b*** can be moved closer together once the artery is in place. The base portions **802*a,b*** can be locked in place when they are sealed together. The base portions **802***a,b* can be crescent shaped components. Together, the base portions **802***a,b* can form a rectangle, for example a rectangle with rounded corners, or an oval. The central aperture **804** can be circular, rectangular, hexagonal, octagonal, or diamond-shaped. The first base portion **802***a* and second base portion **802***b* can include elastic guard portions between them that define the central aperture **804**.

In some examples, a cap portion **806** can be connected to the base portion **802**, for example by a hinge **816**. The cap portion **806** can be sealed on top of the base portion **802** and the vein, artery, or artery cuff. When sealed, the cap portion **806** can be parallel with the base portion **802**. The cap portion **806** can include a barb **808**. The barb **808** can be configured to connect with the cannula receiver **20** of the apparatus **10** of FIG. **1**A-C. The cap portion **806** can include a valve **810**. The valve **810** can be covered, for example by a cap. The valve **810** can be used to deair the cannula once attached, for example using a syringe to remove fluid from the circuit. The cannula **800** can be deaired by connecting a syringe filled with liquid, for example perfusion solution, to the end of the barb **808**, uncapping the valve **810**, and pushing the liquid into the barb **808** until the liquid flows out of the valve **810**. The cannula **800** can be made of plastic, such as copolyester with a silicone seal and gasket. The cap portion **806** can include a small elastic protrusion for keeping the vessel open.

In some examples, the top of the base portion **802** and the bottom of the cap portion **806** can have grooves to better grip the vessel, artery, or artery patch. The grooves on the top of the base portion **802** can align with the grooves on the bottom of the cap portion **806**. The artery patch of the renal artery can be secured between the base portion **802** and the cap portion **806**. In some examples, a flared or stretched out end of the vessel can be secured between the base portion **802** and the cap portion **806**. The barb **808** can have ridges extending radially to lock it into the cannula receiver **20** of the apparatus **10** of FIG. **1**A-C.

In some examples, the cannula **800** can have a clip **812** on the base portion **802**. The clip **812** can mechanically secure the base portion **802** to the cap portion **806** by coupling with a lock portion **814** of the cap portion **806**.

In certain examples, the central aperture **804** can be round or circular to match a lumen of a vein. For example, the round central aperture **804** can match a diameter of a single lumen artery. The round central aperture **804** can have a diameter of approximately 3 mm, 5 mm, 7 mm, or 9 mm. In some embodiments, the round central aperture **804** can have a diameter of approximately 1-20 mm.

FIG. **9**A illustrates an example of an oval-hole cannula **900** in an open configuration. FIG. **9**B illustrates the example of the oval-hole cannula **900** in a closed configuration. The oval-hole cannula **900** can be similar to the cannula **800** of FIGS. **8**A-B.

In some examples, the cannula **900** can be configured to seal to the artery, for example a renal artery. The cannula **900** can include a base portion **902**. A vessel, artery, or artery cuff can be pulled through the central aperture **904**. The base portion can be a rectangle, for example a rectangle with rounded corners, or an oval.

In some examples, a cap portion **906** can be connected to the base portion **902**, for example by a hinge **916**. The cap portion **906** can be sealed on top of the base portion **902** and the vein, artery, or artery cuff. When sealed, the cap portion **906** can be parallel with the base portion **902**. The cap portion **906** can include a barb **908**. The barb **908** can be configured to connect with the cannula receiver **20** of the apparatus **10** of FIG. **1**A-C. The cap portion **906** can include a valve **910**. The valve **910** can be covered, for example by a cap. The valve **910** can be used to deair the cannula once attached, for example using a syringe to remove fluid from the circuit. The cannula **900** can be made of plastic, for example copolyester with a silicone seal and gasket. The cap portion **906** can include a small protrusion for keeping the vessel open.

In some examples, the top of the base portion **902** and the bottom of the cap portion **906** can have grooves to better grip the vessel, artery, or artery patch. The grooves on the top of the base portion **902** can align with the grooves on the bottom of the cap portion **906**. The artery patch of the renal artery can be secured between the base portion **902** and the cap portion **906**. The barb **908** can have ridges extending radially to lock it into the cannula receiver **20** of the apparatus **10** of FIG. **1**A-C.

In certain examples, the cannula **900** can have a clip **912** on the base portion **902**. The clip **912** can mechanically secure the base portion **902** to the cap portion **906** by coupling with a lock portion **914** of the cap portion **906**.

In some examples, the central aperture **904** can be ovular to match a lumen of a vein. For example, the oval central aperture **904** can match a diameter of a double lumen artery. The oval central aperture **904** can have dimensions of approximately 7 mm by 20 mm or 10 mm by 35 mm. In some embodiments, the oval central aperture **904** can have a width of approximately 1-20 mm. In some embodiments, the oval central aperture **904** can have a length of approximately 10-50 mm.

In some examples, once a kidney is recovered, an appropriately sized cannula can be connected to the renal artery. The cannula can be similar to those described with respect to FIGS. **8**A-B or **9**A-B. The cannula can have a clamshell design which is open when connecting the artery and is closed once the renal artery is placed. The artery or patch can be pulled through the aperture **804**, **904** of the cannula using forceps. Once pulled through and seated, the cannula can be closed by snapping the cap portion **806**, **906** and base portion **802**, **902** together. The cannula can be de-aired by flushing cold preservation solution through the cannula and kidney using the barb **808**, **908**, or perfusion port on the cannula. This can be followed by checking for leaks. The kidney can then be placed in the organ container. The canulated kidney can be connected to the fluid circuit within the canister by pressing the valve **810**, **910**, or infusion port, into the cannula receiver. The slider of the cannula receiver can be adjusted up or down as necessary to assure there is little or no tension placed on the renal artery.

FIG. **10** shows an example of an organ retainer **1092**.

The organ retainer **1092** can be similar to the organ retainer in FIG. **7**A. The organ retainer **1092** can be secured to organ posts **1090** using attachment mechanisms **1094**. The organ can be positioned on an organ rest **1018**.

The organ retainer **1092** can include curved members that conform to the shape of the organ. The curved members can be positioned on top of and in contact with the organ to prevent movement of the organ during transportation. In some examples, the organ retainer **1092** can be flexible or elastic. The organ retainer **1092** can deform to maximize contact with the organ. Advantageously, the elasticity of the organ retainer **1092** can allow the organ to be securely retained in place during transportation with reduced risk of damage to the organ.

FIG. **11** is a graph of kidney pressure and vascular resistance in an example of the organ preservation device described herein.

The renal vascular resistances (RVR) shown in the graph are between 0 and 4 mmHg*min/mL. These RVR values are based on 325 deceased donor kidneys at 30 minutes, 1 hour, 2 hours, 4 hours, and at the end of perfusion. At 30 minutes, the mean RVR was 0.28 [0.04, 3.83]. At 30 minutes only three of the 325 kidneys had an RVR above 1.5. At the end of perfusion, the mean RVR was 0.17 [0.02, 1.10].

At a lower RVR, the kidney was observed to be more vasodilated. More flow went through the donor kidney and less flow diverted through the flow regulator valve. The perfusion pressure decreased.

At a higher RVR, the kidney was observed to be constricted. Less flow went through the donor kidney and more flow diverted through the flow regulator valve. The perfusion pressure increased.

FIG. **12** is a graph of kidney pressure and perfusion flow in an example of the organ preservation device described herein.

The perfusion pressure range was between 0 (for an RVR of 0) and 50 mmHg (for a fully occluded donor kidney).

Perfusion pressure was lower as the donor kidney vasodilated. More flow went into the donor kidney and less flow diverted through the flow regulator valve.

Perfusion pressure was higher when the donor kidney was constricted. Less flow went through the kidney and more flow diverted through the flow regulator valve.

FIG. **13**A shows an example of a flow regulator valve **48** in the closed configuration. FIG. **13**B shows the example of the flow regulator valve **48** of FIG. **13**A in the partially open configuration. FIG. **13**C shows the example of the flow regulator valve **48** of FIG. **13**A in the open configuration.

In some examples, the flow regulator valve **48** can mechanically open and close to varying degrees based on the pressure within the circuit. During operation, the flow regulator valve **48** can passively and progressively open and close in direct response to the pressure within the circuit. When pressure within the circuit is higher (due to higher RVR), the flow regulator valve **48** can open, allowing more flow through the flow regulator valve **48** and back into the inner canister. When pressure within the circuit decreases (due to lower RVR), the flow regulator valve **48** can close such that more perfusates passes through the kidney. The flow regulator valve **48** can ensure that pressure within the circuit remains below the perfusion pressure limit, or upper threshold. In some examples, the upper threshold is 65 mmHg. In some examples, the upper threshold is between 50 and 80 mmHg. In some examples, the upper threshold is between 30 and 100 mmHg.

The flow regulator valve **48** can include a ball **1350** and a spring **1352** in a housing **1354**. The ball **1350**, the spring **1352**, and/or the housing can be metal, for example stainless steel. When pressurized in the flow direction F, the spring **1352** can compress and allow fluid to pass around the ball **1350**. When pressure in the flow direction F decreases, the spring **1352** can extend and the ball **1350** can block flow. The properties of the flow regulator valve **48** can be engineered and/or adjusted to have a specific cracking pressure, or the pressure at which the ball is pushed back to allow the fluid to flow through. The flow through the flow regulator valve **48** can be variable as opposed to binary. Advantageously, the flow regulator valve **48** can control flow in a manner directly proportional to the resistance of an artery without requiring active maintenance or software.

FIG. **14** shows a cross-sectional view of an example of an organ preservation apparatus **1400** with a canister bubble trap **1460** and a perfusion line bubble trap **1462**.

The organ preservation apparatus **1400** can be similar to the apparatus **10** of FIG. **1**A-C. The canister **1438** can include a filter **1404**.

The apparatus **1400** can include a canister bubble trap **1460**. The canister bubble trap **1460** can be integrated in the inner lid **1440**. For example, the canister bubble trap **1460** can be integrated in the center of the inner lid **1440**. The canister bubble trap **1460** can prevent air from entering and/or remaining in the canister **1438**.

The apparatus **1400** can include a perfusion line bubble trap **1462**. The perfusion line bubble trap **1462** can be integrated in the inner lid **1440**. For example, the perfusion line bubble trap **1462** can be integrated in the perfusion circuit as fluid is pumped to and/or from the canister **1438**. The perfusion line bubble trap **1462** can prevent air from entering and/or remaining in the perfusion circuit.

FIG. **15** shows an example of an organ preservation apparatus **1500** for preserving an organ.

The organ preservation apparatus **1500** can include any and/or all of the features of the apparatus **10** or the apparatus **1400** as described with respect to FIGS. **1**A-C and **14**.

The apparatus **1500** can be used to ensure or promote homogenous cooling of a donor organ. The apparatus **1500** can circulate cold preservation fluid in a manner that efficiently and effectively cools the organ from within the organ.

In some examples, the organ can be a liver. Because the liver has a larger mass than certain other organs, the liver can be more difficult to cool evenly. In some examples, the organ can be a kidney, a heart, a lung, and/or a pancreas. The organ or biological sample T can be contained in nested containers **1580**. In some examples, the biological sample T can be contained in a bag, a container, or nested bags.

Preservation from inside the nested containers **1580** can flow through the first channel **1582**. The pump **1502** can pump the fluid from inside the nested containers **1580** along the first channel **1582** to a temperature maintenance mechanism **1588**, or heat exchanger. Due to the proximity of the temperature maintenance mechanism **1588** to the cooling media **1586**, the temperature maintenance mechanism **1588** can reduce the temperature of the preservation fluid. In some examples, the temperature maintenance mechanism **1588** can include an active cooling mechanism, for example a refrigerator or an electronic cooler. The fluid can flow from the temperature maintenance mechanism **1588** through the second channel **1584** to a vessel of the organ. For example, the second channel **1584** can connect to a portal vein and/or a hepatic artery of a liver. In some examples, the second channel **1584** can connect to a renal artery of a kidney. The apparatus **1500** can also include cooling media **1586** disposed on or near the lid of the canister.

In some examples, the apparatus **1500** can cool a liver to an approximately homogenous temperature in less than 30 minutes. In some examples, the apparatus **1500** can cool a liver to an approximately homogenous temperature in less than between 15 and 60 minutes. In some examples, the apparatus **1500** can store a liver for approximately 15 hours. In some examples, the apparatus **1500** can store a liver for approximately 5-30 hours.

The cannula coupling the vessel or vessels of the organ with the first channel **1582** can include sensors. For example, the cannula can include a temperature sensor and/or a pressure sensor. The sensors can determine perfusate temperature, portal vein pressure, and/or hepatic artery pressure. The cannula can include a check valve to provide pressure relief to the organ.

FIG. **16** shows an example of transporters **1600** aligned for stacking.

The transporters **1600** can be similar to the transporter **400** described with respect to FIGS. **4**A-**4**B.

The transporters **1600** can be stacked such that protrusions **1690** on the bottom of the transporter **1600** are aligned with and stacked on top of recesses **1692** on the top of a transporter. In some examples, the protrusions **1690** and/or the recesses **1692** can be C-shaped, curved, semicircular, or parenthesis shaped. In some examples, the protrusions **1690** and/or the recesses **1692** can be linear, rectangular, or another shape. In some examples, transporters **1600** can be stacked such that the transporters **1600** are facing the same direction. In some examples, transporters **1600** can be stacked such that the transporters **1600** are facing opposite directions. Advantageously, in organs received in donors who have undergone cardiac death or brain death, two kidneys can be transported together with the transporters **1600** in the stacked formation. This can allow for simultaneous and efficient monitoring of individual organ statistics such as renal pressure and/or temperature.

Stacking transporters **1600** can allow multiple devices to be transported efficiently and securely, which can optimize space during transportation and shipping. Each transporter **1600** can include a wide wheelbase to improve stability and maneuverability of the device. The transporters **1600** can include indents **1696** on or near the front of the transporter **1600**. These indents can cradle the telescoping handle arms of another transporter **1600** to secure the transporters **1600** together when stacked while facing opposite directions. Advantageously, this allows the transporters **1600** to be tilted backwards and rolled together on the wheels of the lower transporter **1600** to enhance mobility and ease of handling during transport. In some examples, each side of the transporter **1600** can include ergonomic areas to handle and maneuver the device, for example indentations on the four faces of the transporter **1600**.

Each transporter **1600** can include a pocket **1698** on the top of the transporter **1600** to contain paperwork. For example, the pocket **1698** can be a clear, plastic pocket. In some examples, the pocket **1698** can be used to store OPO paperwork.

FIG. **17** shows an example of a transporter **1600** aligned for placement in a package **1700**.

The transporter **1600** can have a specialized package **1700** to contain the transporter **1600**. The specialized package **1700** can include recesses **1792** configured to receive the protrusions **1990** of the transporter **1600**. This can simplify the process of removing and replacing the device from the packaging.

FIG. **18**A shows an example of a removable electronics panel **1800**. FIG. **18**B shows the example of the electronics panel of FIG. **18**A being positioned on the transporter **1600**.

The electronics panel **1800** can be removed from the transporter **1600**. The electronics panel **1800** can include a display and a processor. The electronics panel **1800** can be accessible for ease of maintenance and repair.

In some examples, the display can indicate organ perfusion status in real-time, organ ID number, blood type, cross-clamp and total infusion time, perfusate temperature, hepatic and portal flow, renal pressure, pressure, and/or total flow.

EXAMPLES

Additional systems and methods are disclosed in the examples below, which should not be viewed as limiting the invention in any way.

Example 1. A system for hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, comprising: a transport container configured to receive an organ, the transport container configured to be at least partially filled with preservation fluid, the transport container comprising: a fluid circuit comprising: an inlet configured to be in communication with the preservation fluid in the transport container; a flow regulator valve in fluid communication with the inlet, the flow regulator valve configured to release the preservation fluid from the fluid circuit into the transport container at a rate based on vessel resistance of the organ; and an outlet in fluid communication with the flow regulator valve and the organ; and a pump configured to pump the preservation fluid from the inlet to the organ, the pump configured to operate independently of vessel resistance.

Example 2. The system of Example 1, wherein the transport container further comprises an organ rest configured to support the organ.

Example 3. The system of Example 2, further comprising an organ adapter configured to seal a vessel of the organ, the organ adapter adjustable along a groove of the organ rest.

Example 4. The system of any one of Examples 1-3, further comprising a pressure dampener, wherein the pressure dampener is configured to reduce pulsation in the fluid circuit caused by the pump.

Example 5. The system of any one of Examples 1-4, wherein the pump is a peristaltic pump.

Example 6. The system of any one of Examples 1-5, wherein the flow regulator valve releases the preservation fluid from the fluid circuit into the transport container at a higher rate when vessel resistance is higher.

Example 7. The system of any one of Examples 1-6, wherein the transport container further comprises: a plurality of posts; and an organ retainer adjustably couplable to the plurality of posts.

Example 8. The system of any one of Examples 1-7, wherein the transport container further comprises a one-way valve in fluid communication with the fluid circuit, the one-way valve configured to take in preservation fluid from the transport container when fluid is drawn out of the fluid circuit.

Example 9. The system of any one of Examples 1-8, wherein the transport container further comprises a temperature sensor.

Example 10. The system of any one of Examples 1-9, further comprising a lid comprising a fill port and a vent port.

Example 11. The system of Example 10, further comprising an accumulation chamber configured to seal to at least one of the fill port and the vent port, the accumulation chamber comprising a balloon configured to expand when the preservation fluid expands and contract when the preservation fluid contracts.

Example 12. The system of Example 3, wherein the organ adapter comprises: a cannula comprising a barb, the cannula configured to seal to the vessel of the organ; and a cannula receiver configured to couple with the barb.

Example 13. A method for hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, the method comprising: placing an organ in a transport container, the transport container comprising an organ adapter; sealing the organ adapter to a vessel of the organ; placing a lid on the transport container; filling the transport container at least partially with preservation fluid through a fill port in the lid; and activating a pump, the pump configured to pump the preservation fluid from the transport container to a flow regulator valve and the organ adapter, the flow regulator valve configured to release the preservation fluid into the transport container at a rate based on vessel resistance of the organ, and the pump configured to operate independently of vessel resistance.

Example 14. The method of Example 13, further comprising placing the organ on an organ rest.

Example 15. The method of Example 14, further comprising adjusting the organ adapter along a groove of the organ rest.

Example 16. The method of any one of Examples 13-15, further comprising attaching an organ retainer to a plurality posts in the transport container such that the organ retainer secures the organ.

Example 17. The method of any one of Examples 13-16, further comprising sealing an accumulation chamber to at least one of the fill port or a vent port on the lid.

Example 18. A system for hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, comprising: a transport container configured to receive an organ, the transport container configured to be at least partially filled with preservation fluid; a pump configured to pump the preservation fluid from the transport container to the organ in a fluid circuit, the pump configured to operate independently of vessel resistance; and a flow regulator valve in fluid communication with the fluid circuit, the flow regulator valve configured to release the preservation fluid from the fluid circuit into the transport container at a rate based on vessel resistance of the organ.

Example 19. The system of Example 18, wherein the transport container further comprises an organ rest configured to support the organ.

Example 20. The system of Example 19, further comprising an organ adapter configured to seal a vessel of the organ, the organ adapter adjustable along a groove of the organ rest.

Example 21. The system of any one of Examples 18-20, further comprising a pressure dampener, wherein the pressure dampener is configured to reduce pulsation caused by the pump.

Example 22. The system of any one of Examples 18-21, wherein the pump is a peristaltic pump.

Example 23. The system of any one of Examples 18-22, wherein the flow regulator valve releases the preservation fluid from the fluid circuit into the transport container at a higher rate when vessel resistance is higher.

Example 24. The system of any one of Examples 18-23, wherein the transport container further comprises: a plurality of posts; and an organ retainer adjustably couplable to the plurality of posts.

Example 25. The system of any one of Examples 18-24, wherein the transport container further comprises a one-way valve in fluid communication with the fluid circuit, the one-way valve configured to take in preservation fluid from the transport container when fluid is drawn out of the fluid circuit.

Example 26. The system of any one of Examples 18-25, wherein the transport container further comprises a temperature sensor.

Example 27. The system of any one of Examples 18-26, further comprising a lid comprising a fill port and a vent port.

Example 28. The system of Example 27, further comprising an accumulation chamber configured to seal to at least one of the fill port and the vent port, the accumulation chamber comprising a balloon configured to expand when the preservation fluid expands and contract when the preservation fluid contracts.

Example 29. The system of Example 20, wherein the organ adapter comprises: a cannula comprising a barb, the cannula configured to seal to the vessel of the organ; and a cannula receiver configured to couple with the barb.

Example 30. A system for hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, comprising: a canister configured to contain an organ and preservation fluid; an inner lid configured to couple with the canister; an outer canister configured to couple with the inner lid, the outer canister comprising a tube; and a pump configured to pump the preservation fluid in the tube such that the preservation fluid flows from the canister to the inner lid, from the inner lid to the outer canister, from the outer canister to the tube, from the tube to the outer canister, from the outer canister to the inner lid, from the inner lid to the organ, and from the organ to the canister.

Example 31. A system for hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, comprising: a canister configured to receive an organ, the canister containing preservation fluid; a port between the canister and an environment; and an accumulation chamber configured to seal to the port, the accumulation chamber comprising a balloon configured to expand when the preservation fluid expands and contract when the preservation fluid contracts.

Example 32. A method for hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, the method comprising: placing an organ in a canister; filling the canister with preservation fluid through a fill port; venting the canister of fluid through a vent port; and attaching an accumulation chamber on at least one of the fill port or the vent port, the accumulation chamber comprising a balloon configured to expand when the preservation fluid expands and contract when the preservation fluid contracts.

Example 33. A system for hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, comprising: a canister configured to receive an organ, the canister comprising: an organ rest configured to support the organ; and a plurality of posts; and an organ retainer adjustably couplable to the plurality of posts.

Example 34. A method for hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, the method comprising: placing an organ in a canister, the canister comprising an organ rest configured to support the organ and a plurality of posts; and coupling an organ retainer to the plurality of posts at a position such that the organ retainer contacts the organ.

Example 35. A system for hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, comprising: a canister configured to receive an organ; an organ rest inside the canister configured to support the organ, the organ rest comprising a groove; and an organ adapter configured to seal to a vessel of the organ, the organ adapter movable along the groove.

Example 36. The system of Example 35, wherein the organ adapter comprises: a cannula configured to seal to the vessel of the organ; and a cannula receiver on the organ rest configured to couple with the cannula.

Example 37. A method for hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, the method comprising: placing an organ in a canister on an organ rest, the organ rest comprising a groove; sealing an organ adapter to a vessel of the organ; moving the organ adapter along the groove; and locking the organ adapter in place such that a vessel artery of the organ is under tension.

Example 38. The method of Example 37, wherein sealing the organ adapter to the vessel of the organ comprises:

sealing the vessel of the organ with a cannula; and coupling the cannula to a cannula receiver on the organ rest.

Example 39. A system for hypothermic transport of an organ, for example a kidney, a lung, a heart, or a liver, comprising: a transport container configured to receive an organ, the transport container configured to be at least partially filled with preservation fluid, the transport container comprising: a fluid circuit comprising: an inlet configured to be in communication with the preservation fluid in the transport container; a heat exchanger in fluid communication with the inlet, the heat exchanger configured to cool the preservation fluid in the fluid circuit; and an outlet in fluid communication with the heat exchanger and a vessel of the organ; and a pump configured to pump the preservation fluid from the inlet to the heat exchanger and from the heat exchanger to the organ.

Example 40. The system of Example 39, wherein the transport container further comprises an organ rest configured to support the organ.

Example 41. The system of Example 40, further comprising an organ adapter configured to seal a vessel of the organ, the organ adapter adjustable along a groove of the organ rest.

Example 42. The system of any one of Examples 39-41, further comprising a pressure dampener, wherein the pressure dampener is configured to reduce pulsation in the fluid circuit caused by the pump.

Example 43. The system of any one of Examples 39-42, wherein the pump is a peristaltic pump.

Example 44. The system of any one of Examples 1-12, further comprising a bubble trap integrated in a lid of the transport container.

Example 45. The system of any one of Examples 1-12, further comprising

a bubble trap integrated in the fluid circuit.

Example 46. An apparatus substantially as shown and/or described.

Example 47. A method substantially as shown and/or described.

Example 48. A system substantially as shown and/or described.

In various embodiments, cooling blocks may include eutectic cooling media or other phase change material (PCM) such as savENRG packs with PCM HS01P material commercially available from RGEES, LLC or Akuratemp, LLC (Arden, NC). Exemplary PCM specifications including a freezing temperature of 0° C.+/0.5° C., a melting temperature of 1° C.+/0.75° C., latent heat of 310 J/g+/10 J/g, and density of 0.95 gram/ml+/0.05 gram/ml. Pouch dimensions may vary depending on application specifics such as tissue to be transported and the internal dimensions of the transport container and external dimensions of the tissue storage device, chamber, or canister. PCM may be included in pouches approximately 10 inches by 6 inches having approximately 230 g of PCM therein. Pouches may be approximately 8.5 mm thick and weigh about 235 g to 247 g. In some embodiments, pouches may be approximately 6.25 inches by 7.75 inches with a thickness of less than about 8.5 mm and a weight of between about 193 g and about 201 g. Other exemplary dimensions may include about 6.25 inches by about 10 inches. Pouches may be stacked or layered, for example in groups of 3 or 4 to increase the total thickness and amount of PCM. In certain embodiments, PCM containing pouches may be joined side to side to form a band of coupled PCM pouches. Such a band may be readily manipulated to wrap around the circumference of a cylindrical storage container and may have dimensions of about 6 inches by about 26 inches consisting of approximately 8 individual pouches joined together in the band.

A variety of preservation solutions can be used with the disclosed systems, devices, and methods. This includes approved preservation solutions, such as Histidine Tryptophan Ketoglutarate (HTK) (e.g., HTK Custodial™) and Celsior™ solutions for the preservation of hearts and cardiac tissues, and University of Wisconsin Solution (Viaspan™) and MPS **1** for the preservation of kidney and kidney tissues. Other preservation solutions, including non approved solutions, and off label applications of approved solutions can be used with the devices described herein. Various preservation solutions can be used, including Collins, EuroCollins, phosphate buffered sucrose (PBS), University of Wisconsin (UW) (e.g., Belzer Machine Preservation Solution (MPS)), histidine tryptophan ketoglutarate (HTK), hypertonic citrate, hydroxyethyl starch, and Celsior™. Additional details of these solutions can be found at t'Hart et al. "New Solutions in Organ Preservation," Transplantation Reviews 2006, vol. 16, pp. 131 141 (2006).

Various modifications to the implementations described in this disclosure may be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other implementations without departing from the spirit or scope of this disclosure. Thus, the disclosure is not intended to be limited to the implementations shown herein, but is to be accorded the widest scope consistent with the principles and features disclosed herein. Various combinations and subcombinations of the various features described herein are possible. Certain embodiments are encompassed in the claim set listed below.

Although this disclosure describes certain embodiments, it will be understood by those skilled in the art that many aspects of the methods and devices shown and described in the present disclosure may be differently combined and/or modified to form still further embodiments or acceptable examples. All such modifications and variations are intended to be included herein within the scope of this disclosure. Indeed, a wide variety of designs and approaches are possible and are within the scope of this disclosure. No feature, structure, or step disclosed herein is essential or indispensable. Moreover, while illustrative embodiments have been described herein, the scope of any and all embodiments having equivalent elements, modifications, omissions, combinations (e.g., of aspects across various embodiments), substitutions, adaptations and/or alterations as would be appreciated by those in the art based on the present disclosure. While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of protection.

Features, materials, characteristics, or groups described in conjunction with a particular aspect, embodiment, or example are to be understood to be applicable to any other aspect, embodiment or example described in this section or elsewhere in this specification unless incompatible therewith. All of the features disclosed in this specification (including any accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. The protection is not restricted to the details of any foregoing embodiments. The protection extends to any novel one, or any novel combination, of the features disclosed in this specification (including any accompanying claims, abstract and drawings), or to any novel one, or any novel combination, of the steps of any method or process so disclosed.

Furthermore, certain features that are described in this disclosure in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations, one or more features from a claimed combination can, in some cases, be excised from the combination, and the combination may be claimed as a subcombination or variation of a subcombination.

Moreover, while operations may be depicted in the drawings or described in the specification in a particular order, such operations need not be performed in the particular order shown or in sequential order, or that all operations be performed, to achieve desirable results. Other operations that are not depicted or described can be incorporated in the example methods and processes. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the described operations. Further, the operations may be rearranged or reordered in other implementations. Those skilled in the art will appreciate that in some embodiments, the actual steps taken in the processes illustrated and/or disclosed may differ from those shown in the figures. Depending on the embodiment, certain of the steps described above may be removed, others may be added. Furthermore, the features and attributes of the specific embodiments disclosed above may be combined in different ways to form additional embodiments, all of which fall within the scope of the present disclosure. Also, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described components and systems can generally be integrated together in a single product or packaged into multiple products.

For purposes of this disclosure, certain aspects, advantages, and novel features are described herein. Not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, for example, those skilled in the art will recognize that the disclosure may be embodied or carried out in a manner that achieves one advantage or a group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

Conditional language, such as "can," "could," "might," or "may," unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements, and/or steps are included or are to be performed in any particular embodiment.

Conjunctive language such as the phrase "at least one of X, Y, and Z," unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require the presence of at least one of X, at least one of Y, and at least one of Z.

Language of degree used herein, such as the terms "approximately," "about," "generally," and "substantially" as used herein represent a value, amount, or characteristic close to the stated value, amount, or characteristic that still performs a desired function or achieves a desired result. For example, the terms "approximately", "about", "generally," and "substantially" may refer to an amount that is within less than 10% of, within less than 5% of, within less than 1% of, within less than 0.1% of, and within less than 0.01% of the stated amount.

The scope of the present disclosure is not intended to be limited by the specific disclosures of preferred embodiments in this section or elsewhere in this specification, and may be defined by claims as presented in this section or elsewhere in this specification or as presented in the future. The language of the claims is to be interpreted broadly based on the language employed in the claims and not limited to the examples described in the present specification or during the prosecution of the application, which examples are to be construed as non-exclusive.

What is claimed is:

1. A method for hypothermic transport of a biological sample, the method comprising:

sealing a biological sample adapter to a vessel of a biological sample;

placing the biological sample in a transport container;

sealing a lid on the transport container;

filling the transport container at least partially with preservation fluid through a fill port in the lid; and

activating a pump,

the pump configured to pump the preservation fluid through a fluid circuit from the transport container to a flow regulator valve and the biological sample adapter,

the flow regulator valve configured to regulate a flow rate into the biological sample based on at least one biological sample parameter, wherein the flow regulator valve is configured to release fluid from the fluid circuit at a rate based on the at least one biological sample parameter, and the pump configured to operate independently of the at least one biological sample parameter and;

wherein the at least one biological sample parameter comprises at least one of vessel resistance, biological sample temperature, or flow rate through the biological sample.

2. The method of claim **1**, further comprising placing the biological sample on a biological sample rest.

3. The method of claim **2**, further comprising coupling the biological sample adapter to a cannula receiver on the biological sample rest.

4. The method of claim **3**, further comprising adjusting the biological sample adapter along a groove of the biological sample rest.

5. The method of claim **4**, further comprising locking the biological sample adapter in place along the groove such that the vessel of the biological sample is under tension.

6. The method of claim **1**, further comprising attaching a biological sample retainer to a plurality posts in the transport container such that the biological sample retainer secures the biological sample.

7. The method of claim **1**, further comprising venting the transport container of fluid through a vent port.

8. The method of claim **7**, further comprising attaching an accumulation chamber on at least one of the fill port or the vent port, the accumulation chamber comprising a balloon configured to expand when the preservation fluid expands and contract when the preservation fluid contracts.

9. The method of claim 1, further comprising preserving the biological sample for greater than 5 hours without adjusting parameters of the pump.

10. A method for hypothermic transport of a biological sample, the method comprising:

pumping, using a pump, preservation fluid from a transport container to an inlet of a fluid circuit at a rate independent of vessel resistance of a biological sample;

pumping, using the pump, the preservation fluid from the inlet to a flow regulator valve;

regulating, using the flow regulator valve, a flow rate into the biological sample based on at least one biological sample parameter, wherein the flow regulator valve is configured to release fluid from the fluid circuit at a rate based on the at least one biological sample parameter; and

pumping, using the pump, the preservation fluid from the flow regulator valve to a biological sample adapter in fluid communication with the biological sample and;

wherein the at least one biological sample parameter comprises at least one of vessel resistance, biological sample temperature, or flow rate through the biological sample.

11. The method of claim 10, further comprising applying tension to a vessel of the biological sample with the biological sample adapter while the biological sample is on a biological sample rest.

12. The method of claim 10, further comprising expanding a balloon in an accumulation chamber when the preservation fluid expands, wherein the accumulation chamber is attached to a port of a lid of the transport container.

13. The method of claim 10, further comprising cooling the biological sample with phase change material on an outer lid sealed to the transport container.

14. The method of claim 10, further comprising reducing pulsation in the fluid circuit with a pressure dampener.

15. The method of claim 10, further comprising measuring a pressure in the fluid circuit.

16. The method of claim 15, further comprising displaying the pressure in the fluid circuit on a display of the transport container.

17. The method of claim 10, further comprising measuring a temperature in the fluid circuit.

18. The method of claim 17, further comprising displaying the temperature in the fluid circuit on a display of the transport container.

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